Jorge.ripoll@uc3m.es

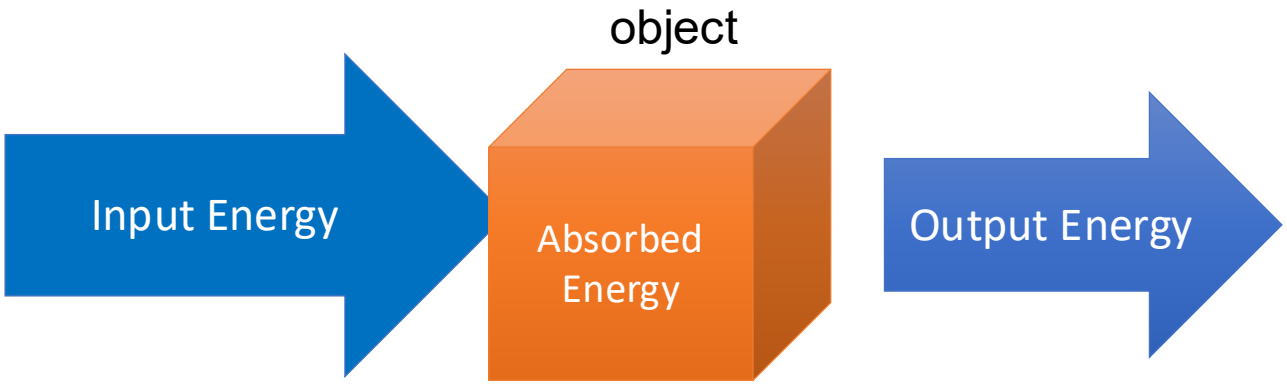
- **Part 1: Optical Imaging**
  - Generation of light
  - Plane wave propagation: geometrical optics
  - Optical Imaging modalities
    - Low scattering (microscopy, OPT, SPIM)
    - High scattering (DOT,FMT)
- **Part 3: Neuroimaging**
  - **Functional MRI**
  - **Functional Diffuse Optical Tomography**
- **Part 5: Biomedical Optics**
- **Part 5: Biomedical analysis technologies**
  - Elisa
  - FACS



OVERVIEW

Scattering, Absorption, Fluorescence and Applications

The Basics: Energy Conservation

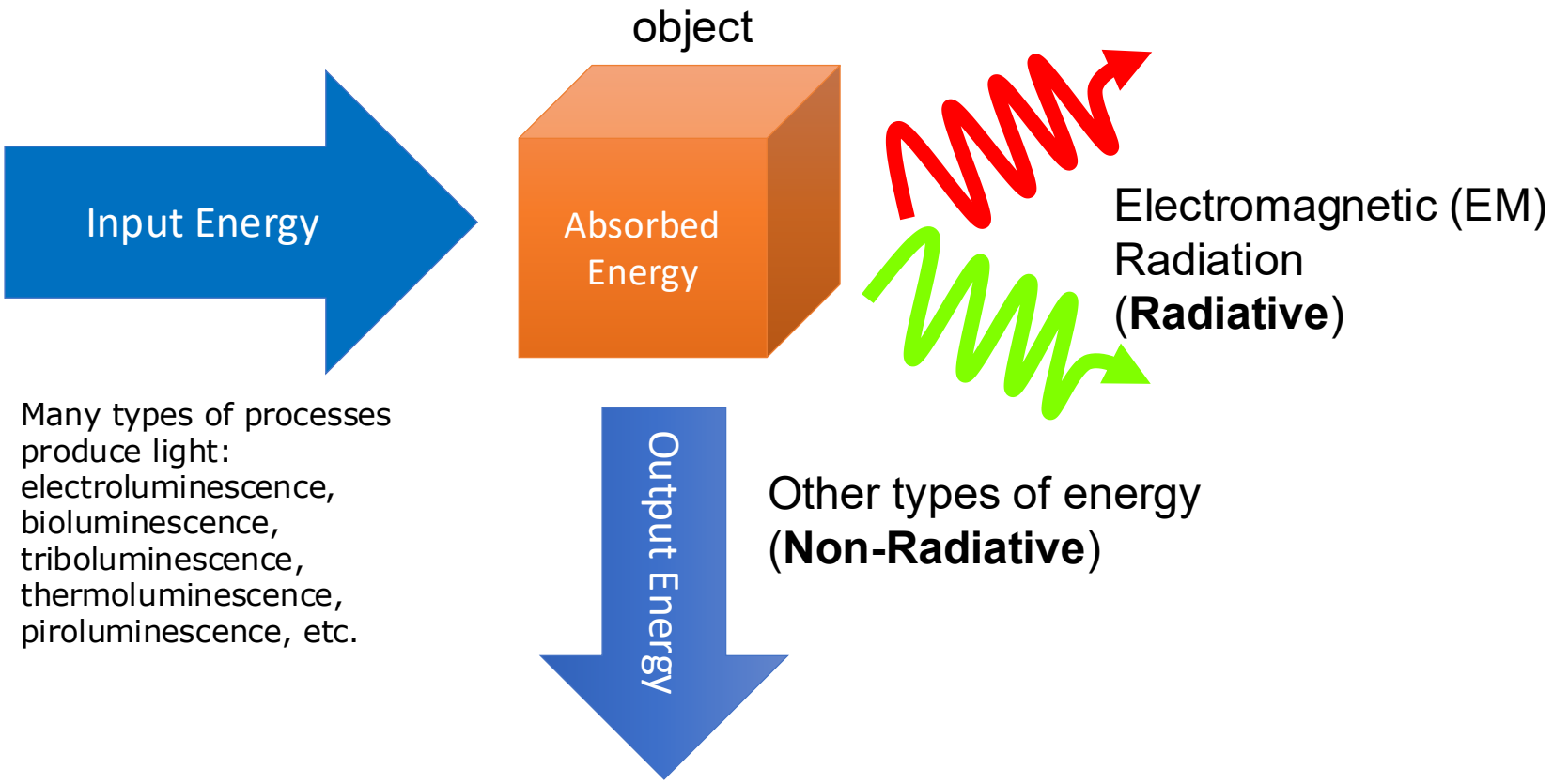


This interaction can have many forms, as many as different forms of energy exist (thermal, electrical, chemical, electromagnetic, kinetic, magnetic, mechanical, nuclear or any combination) . What always must hold true is:

Total Input Energy = Total Energy Absorbed by Object + Total Output Energy

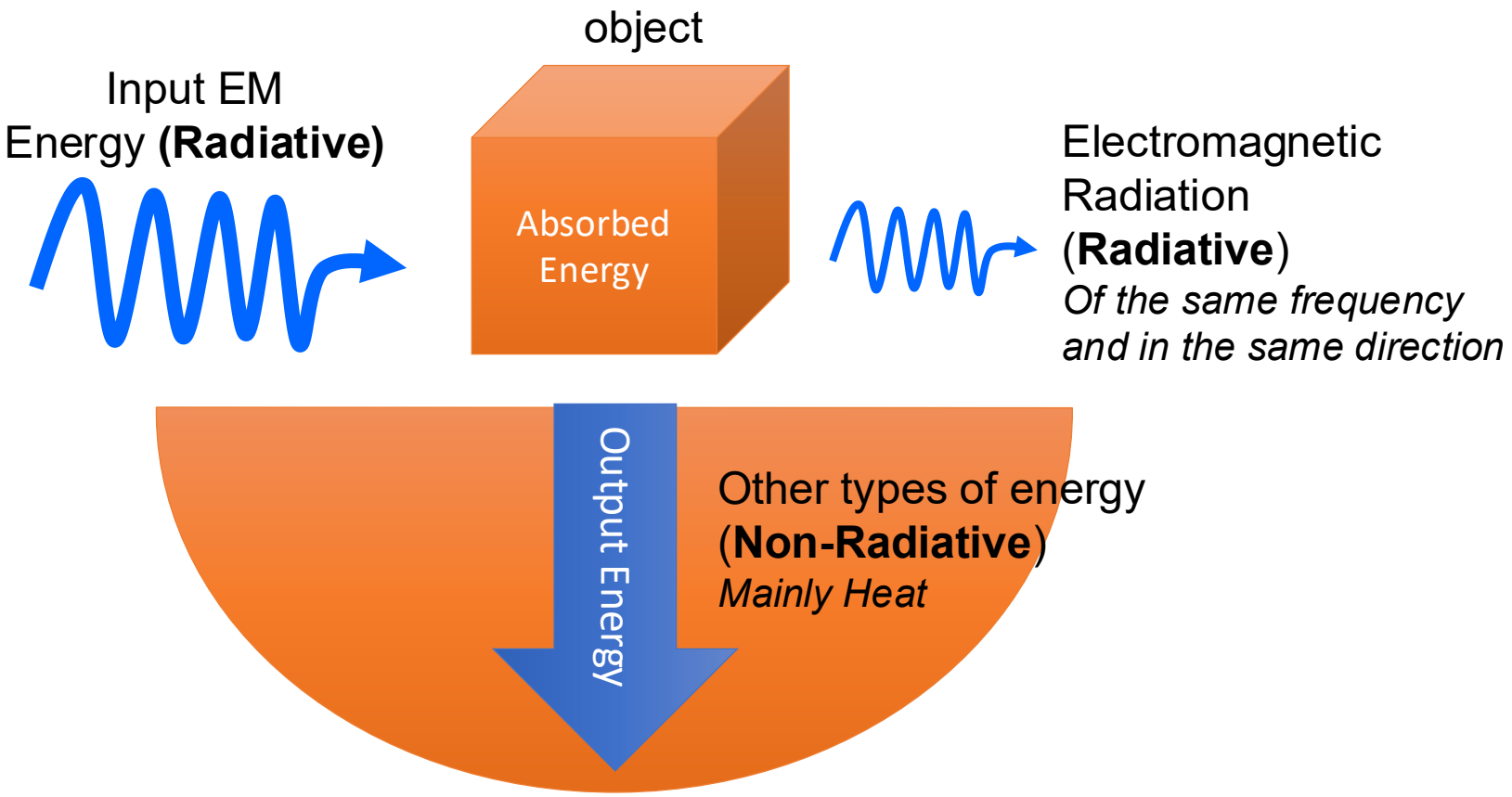
So, all color we see is due to

I) Light Emission



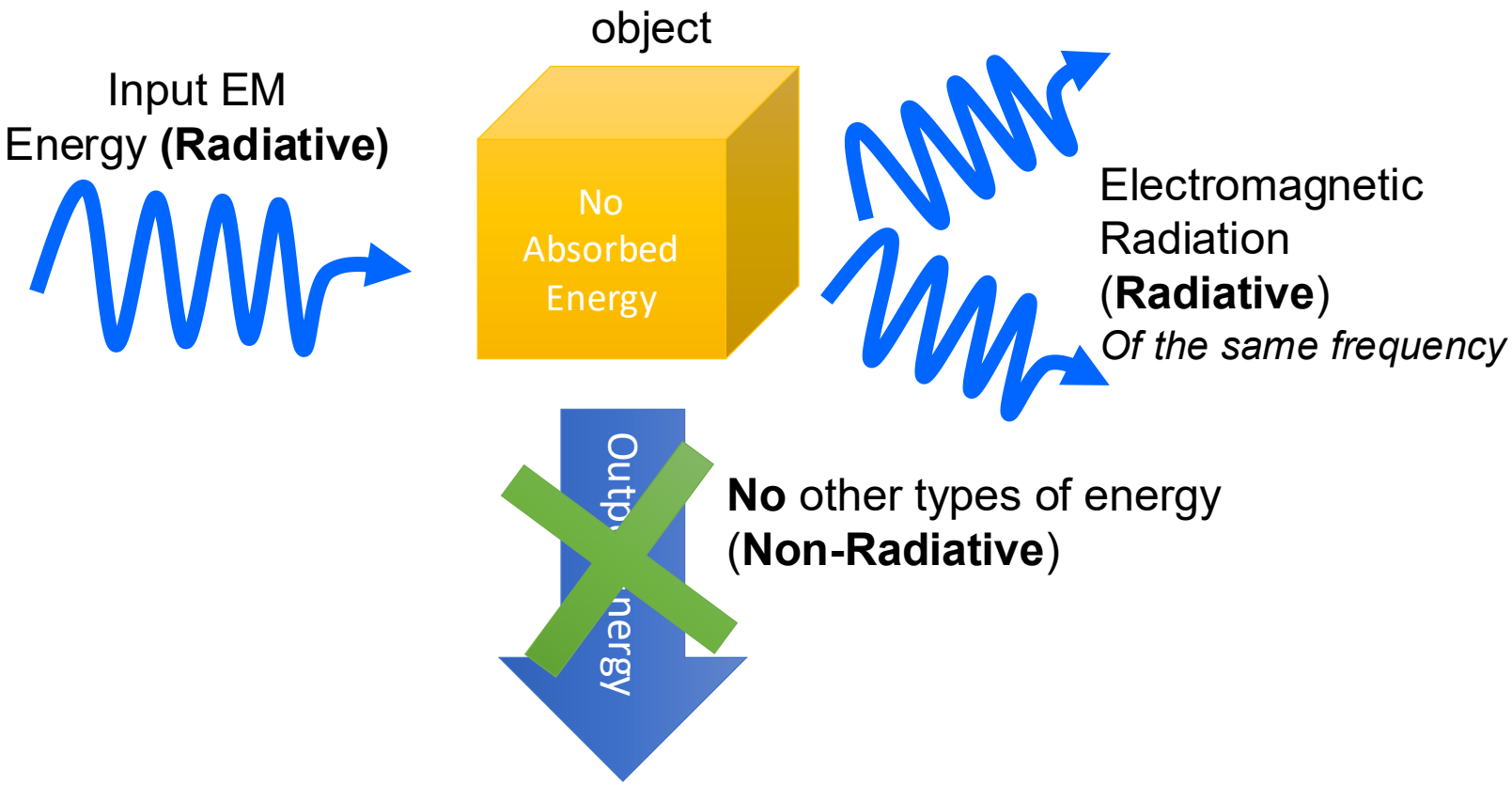
and

II) Light Absorption

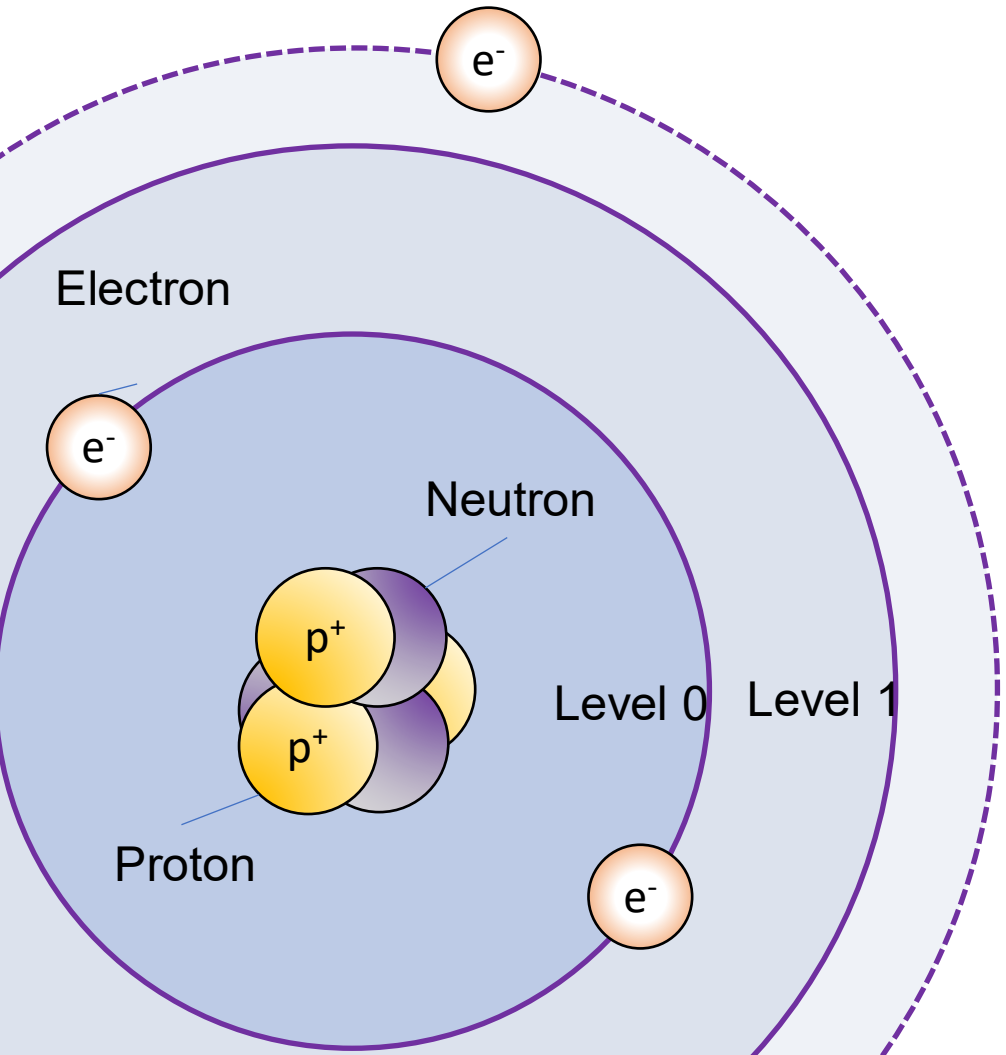


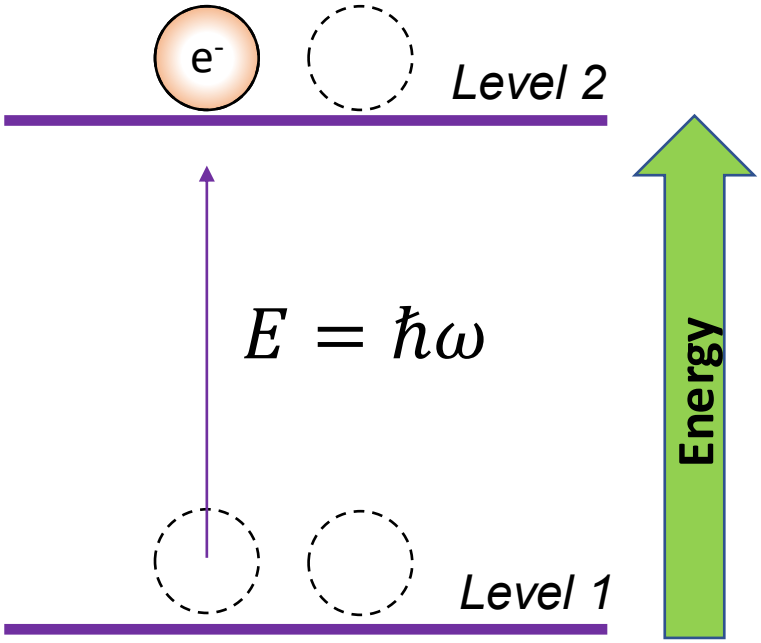
and finally

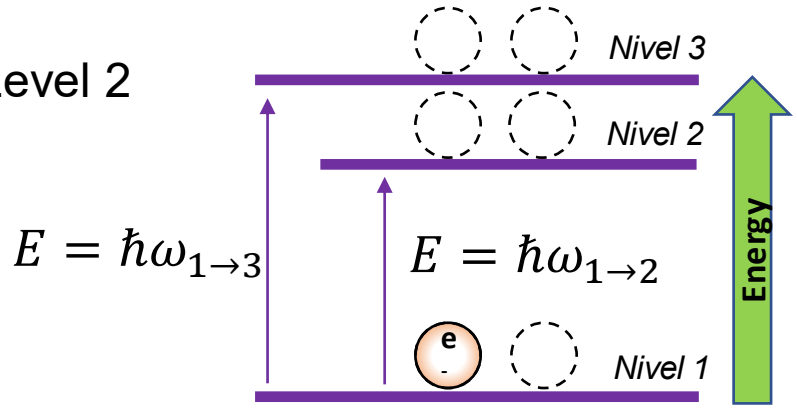
III) Light Scattering



Atomic levels

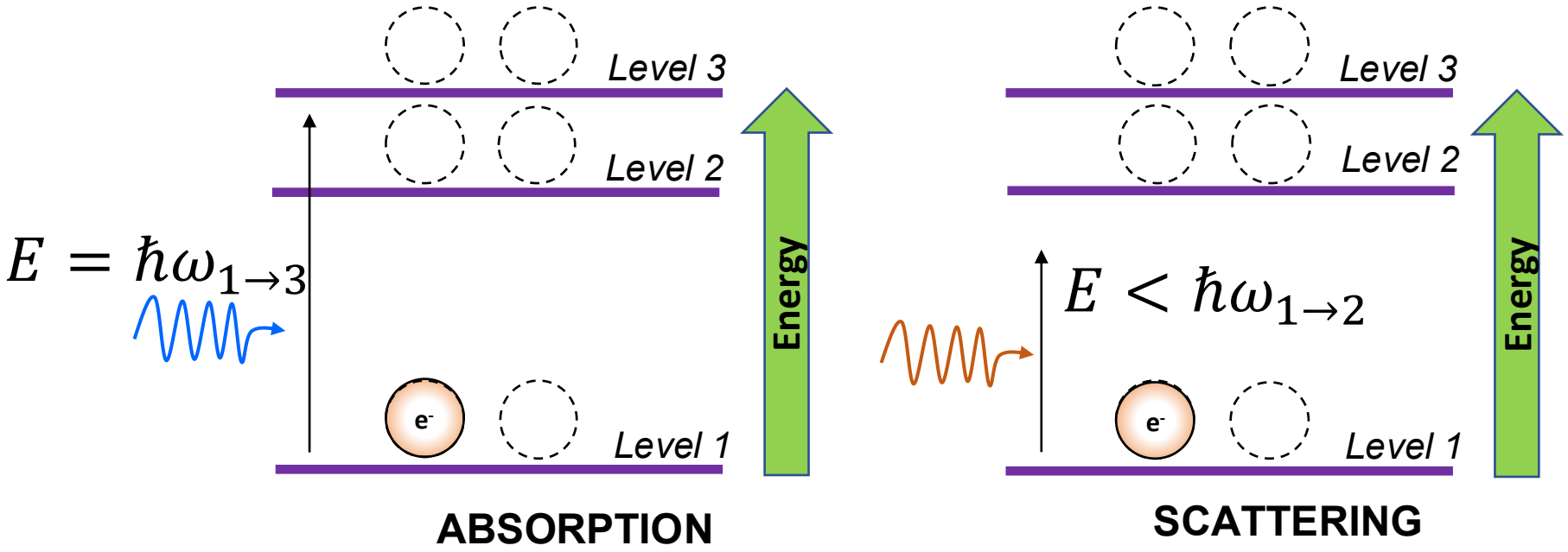






Atomic levels

- Each molecule has its own energy levels
- Each energy level has an associated frequency
- To promote an electron at that level of excitement, the energy provided has to be $E = \hbar\omega$



Optical Properties depend on the collective properties:

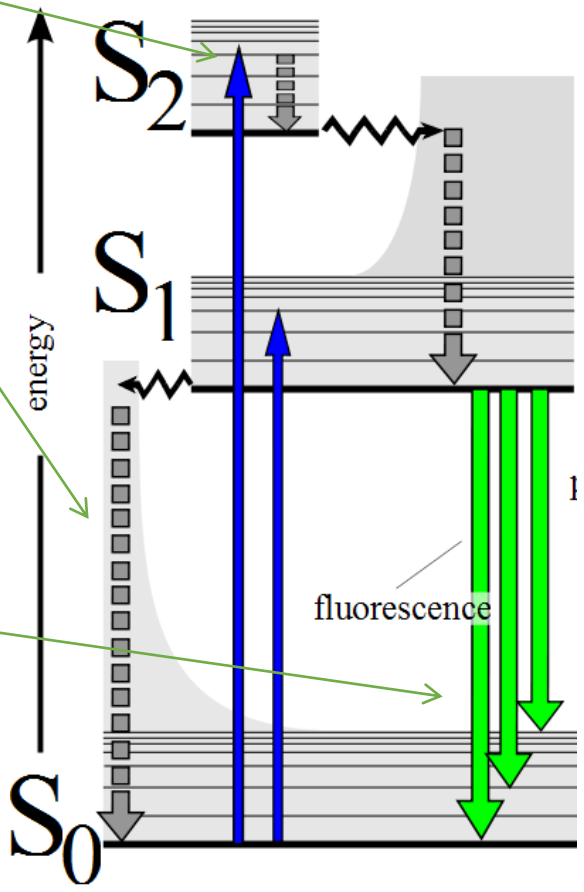
Depending on the levels (orbitals) accessible, this extra energy can be given back to the system either as **non-radiative** or **radiative** emission. It is the radiative emission we're interested in right now.

We managed to excite the electron to this higher energy level (in this case, a singlet state)

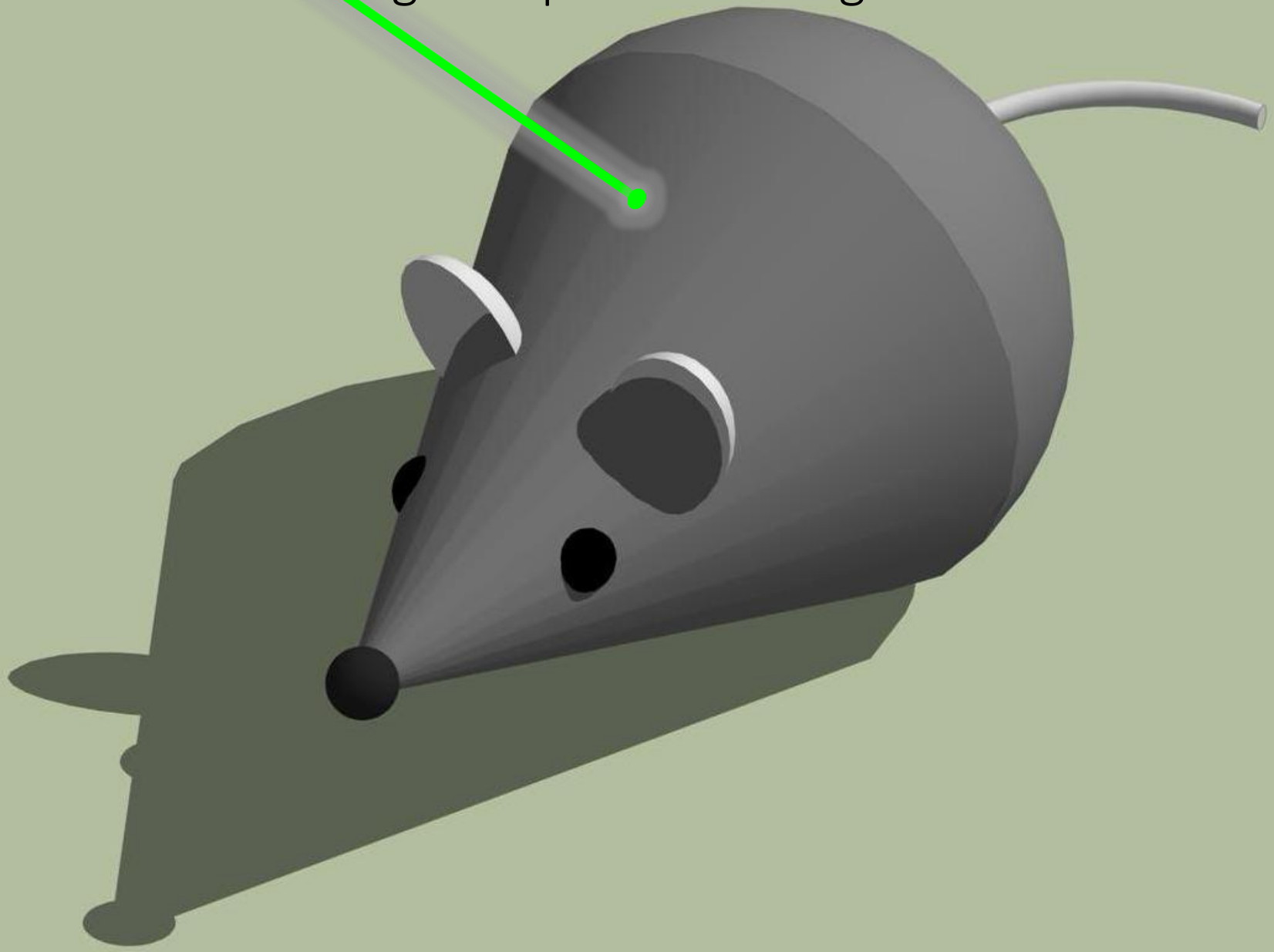
It can then return to the ground (state of minimum energy) either non-radiatively (heating up its environment, for example)

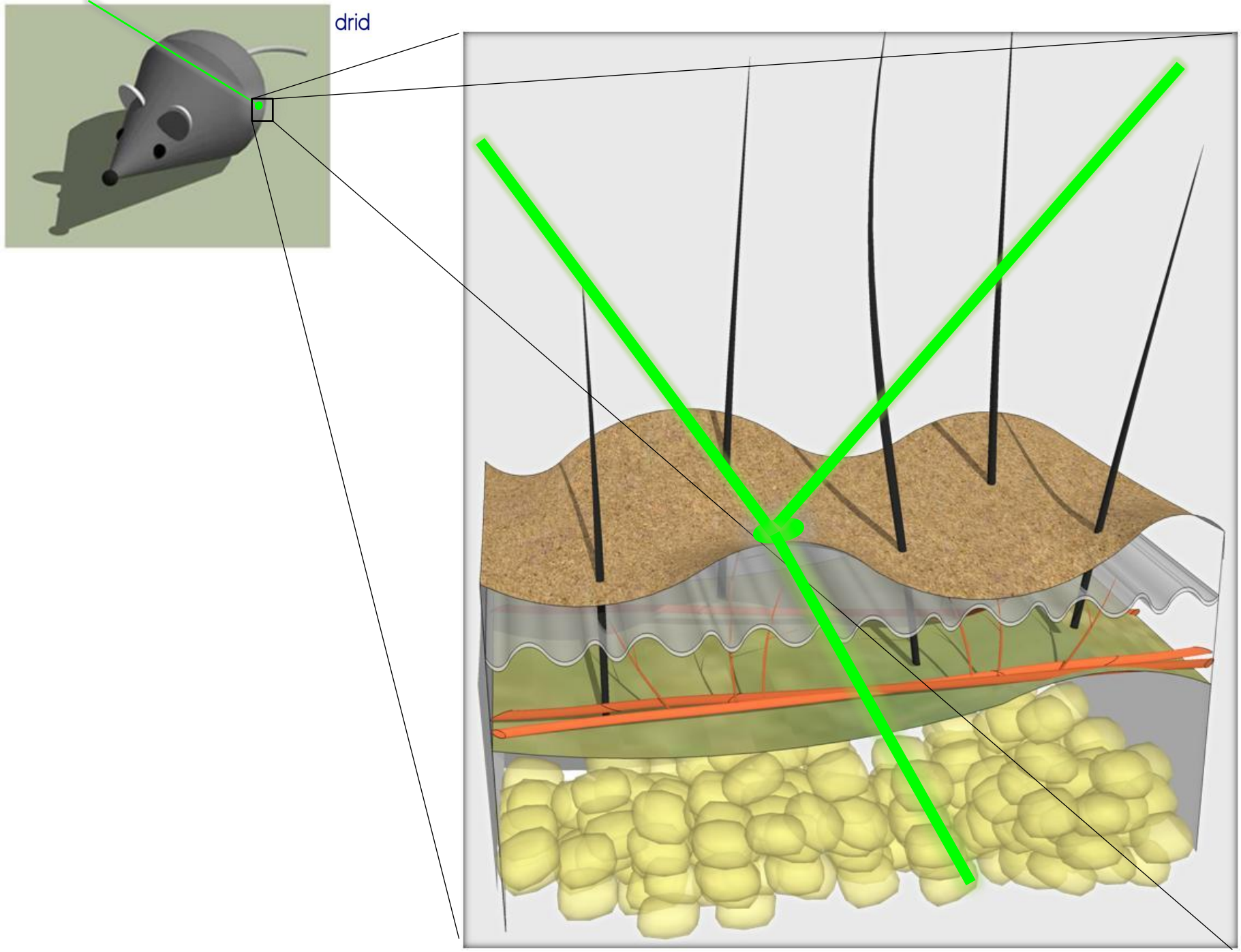
Or radiatively, i.e. emitting EM radiation, as in Fluorescence or Phosphorescence.

Jablonski Diagram

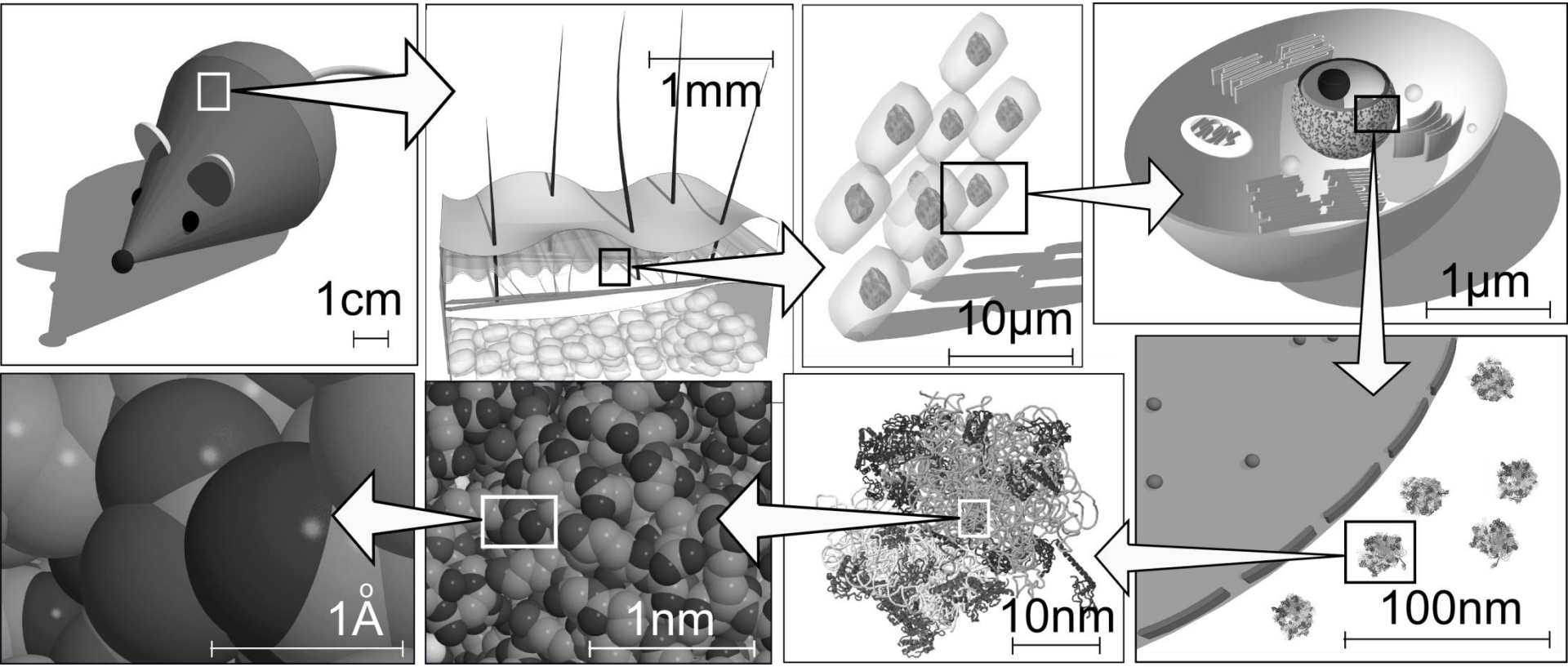


We wish to use light to probe or image tissue



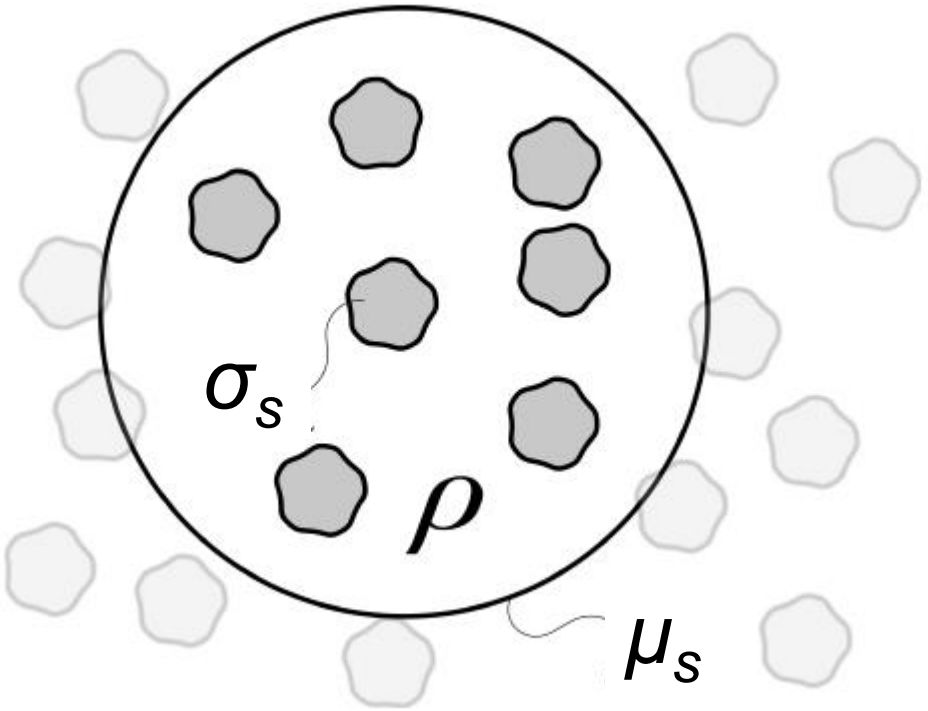

drid

Characteristic Sizes



Scattering

Statistical Description of Optical Properties



In the case where we have a collection of these particles at a certain density, the amount of scattering and absorption will depend on the density of these.

Scattering Coefficient

$$\mu_s = \rho\sigma_{sc}, \qquad \left(cm^{-1}\right)$$

Scattering Anisotropy

We know that the energy is scattered, but we need information on *how* this energy is scattered, i.e. if there is a preferential direction of scattering

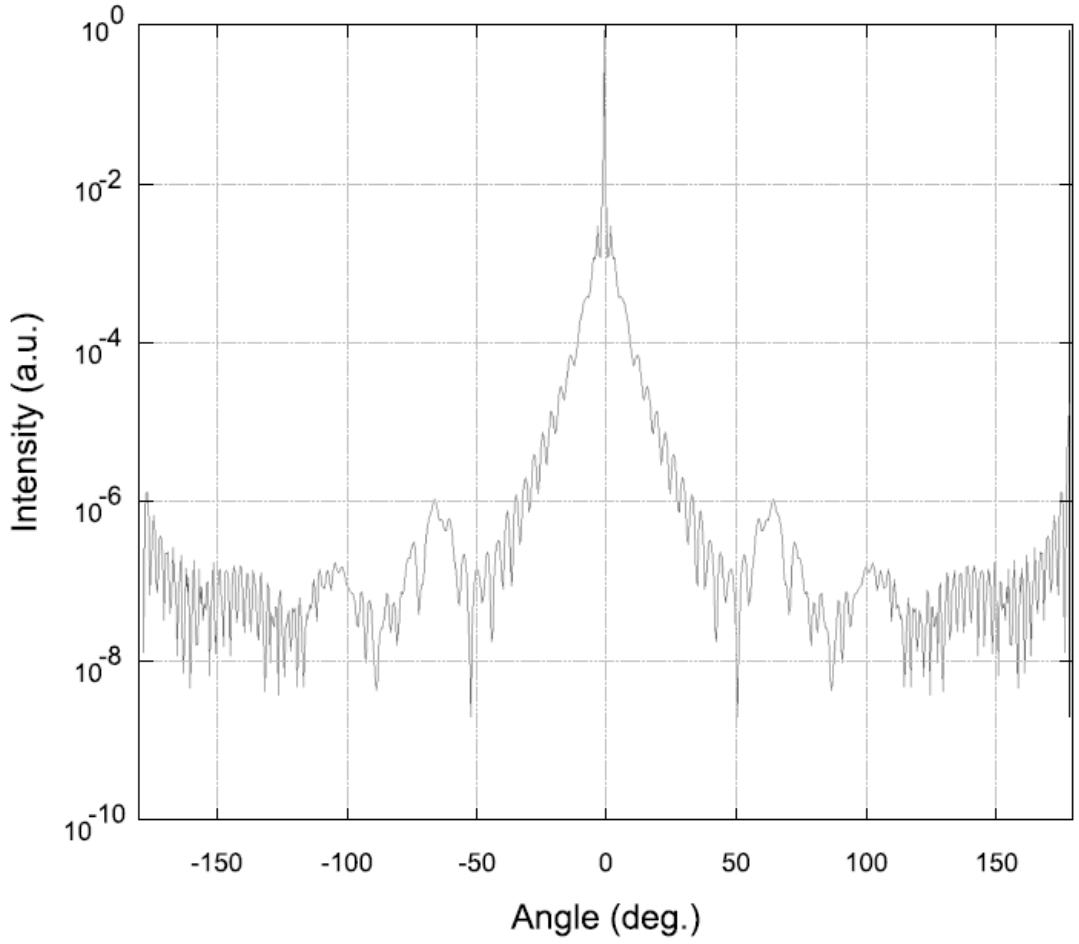


Fig. 1.13 Approximation of what the angular intensity distribution would look like for a fat cell, assuming it is a sphere of radius 50 times the wavelength of the incident light (i.e. 80μm diameter for a 800nm incident wavelength) with index of refraction 1.4 surrounded by water.

Scattering Anisotropy

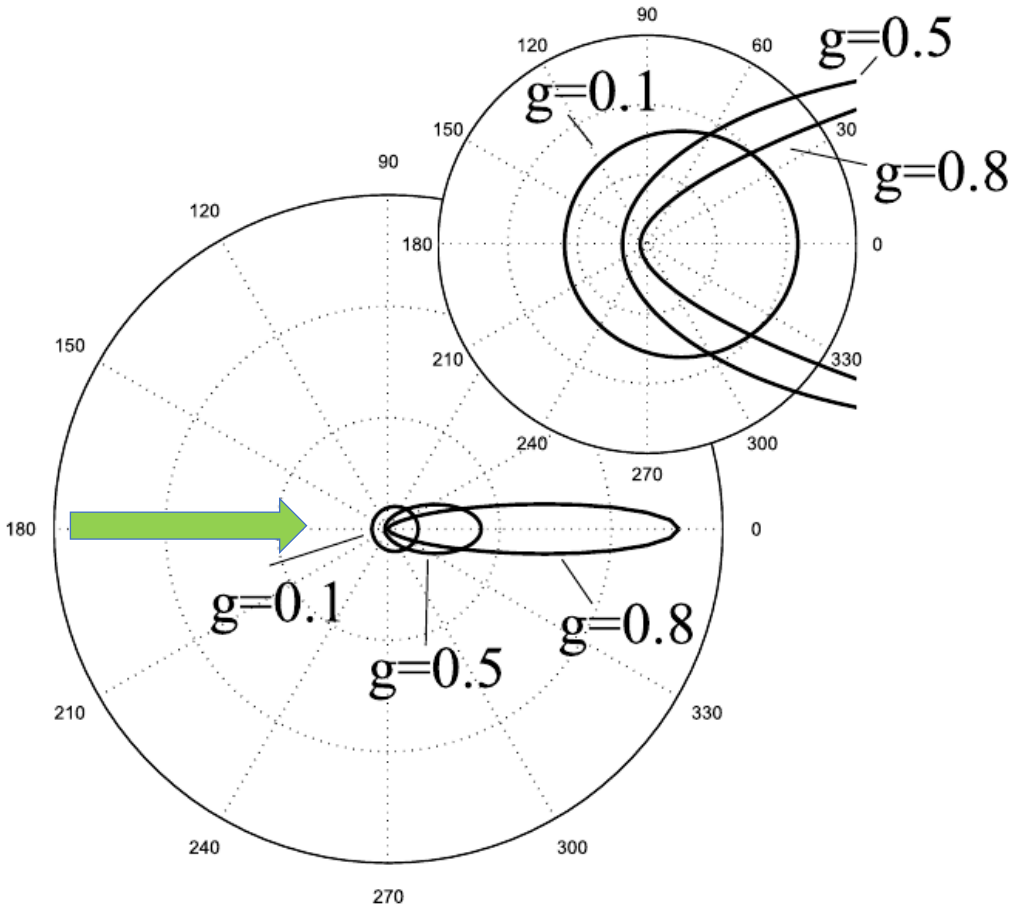
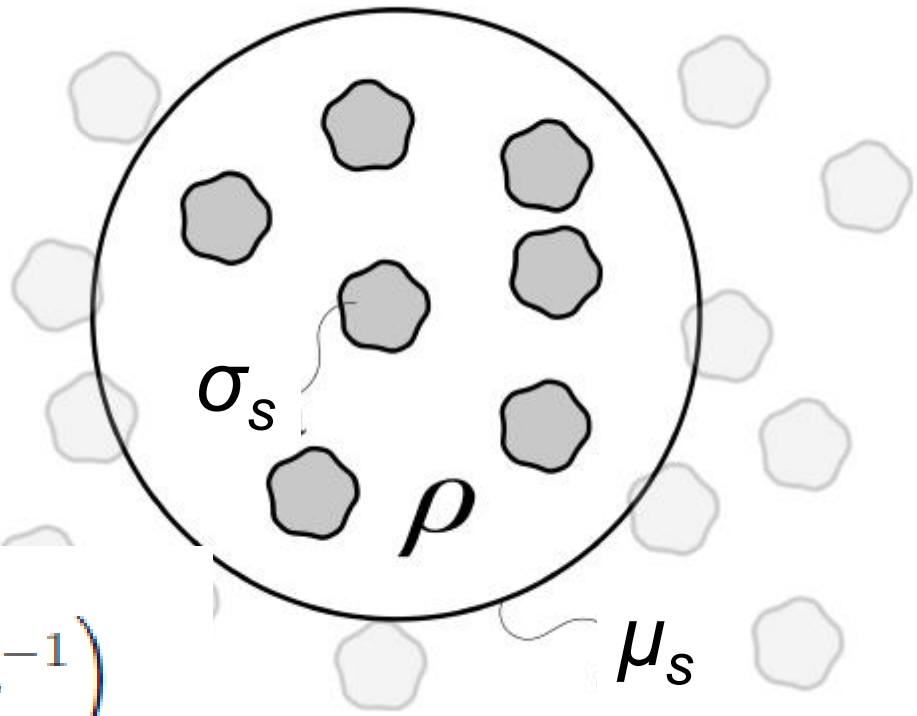


Fig. 1.6 Polar plot of the scattering diagram or phase function for the Henyey-Greenstein expression for anisotropy values of $g = 0.1$, $g = 0.5$ and $g = 0.8$. The inset shows in greater detail the difference for backscattering angles (note that light is incident from the left).

Scattering Anisotropy



Reduced Scattering Coefficient

$$\mu_s' = \mu_s(1-g). \qquad \left(cm^{-1}\right)$$

- The *scattering mean free path*:

$$l_{sc} = \frac{1}{\mu_s}, \qquad \left(cm\right)$$

- and the *reduced scattering mean free path*:

$$l_{sc}' = \frac{1}{\mu_s'} = \frac{l_{sc}}{(1-g)}, \qquad \left(cm\right)$$

Milk Experiment

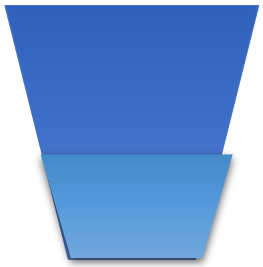
For 1cm depth:



100% Milk



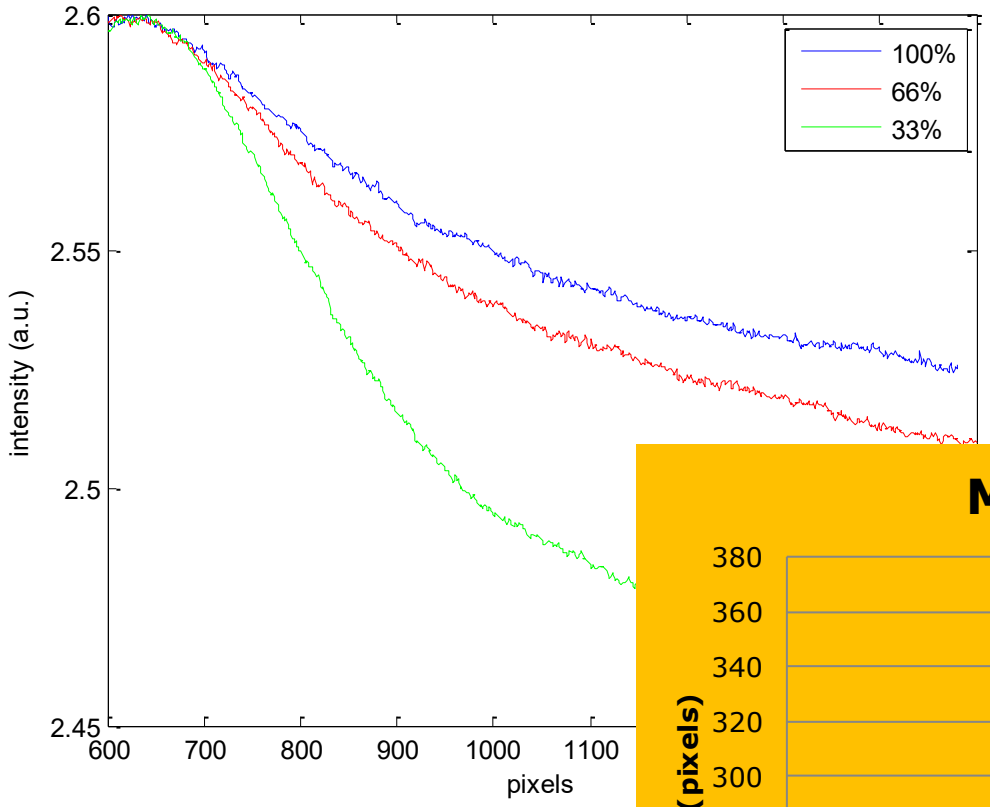
66% Milk

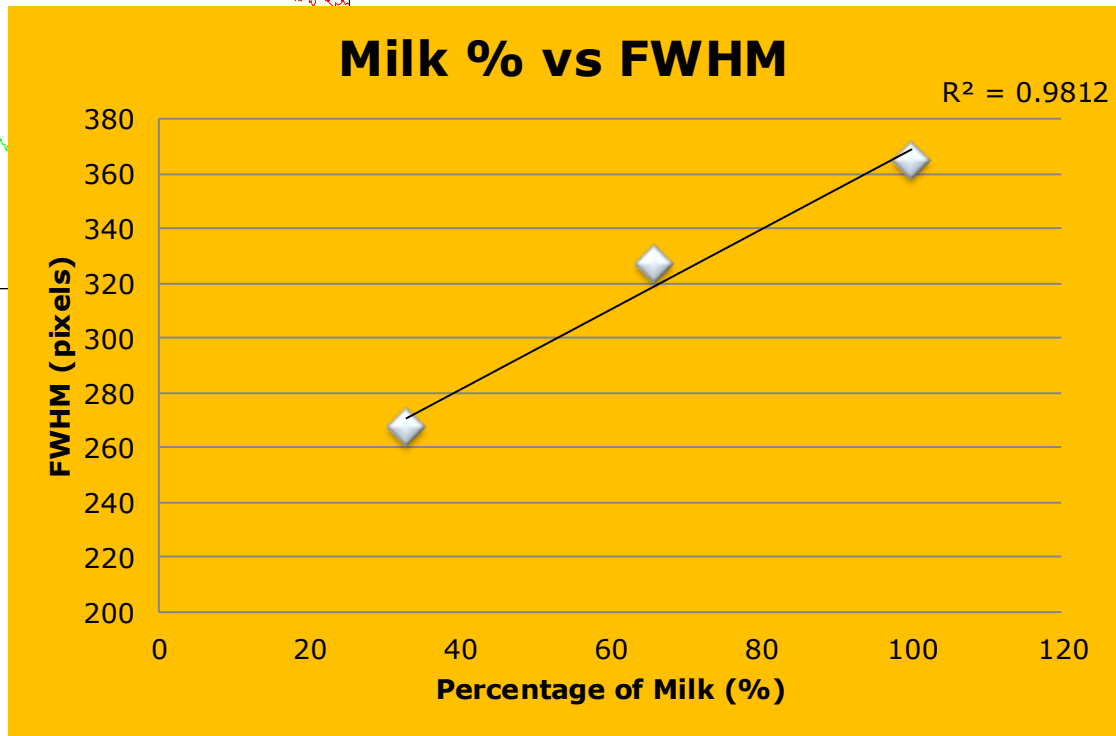


33% Milk



Analysis: 100% Milk

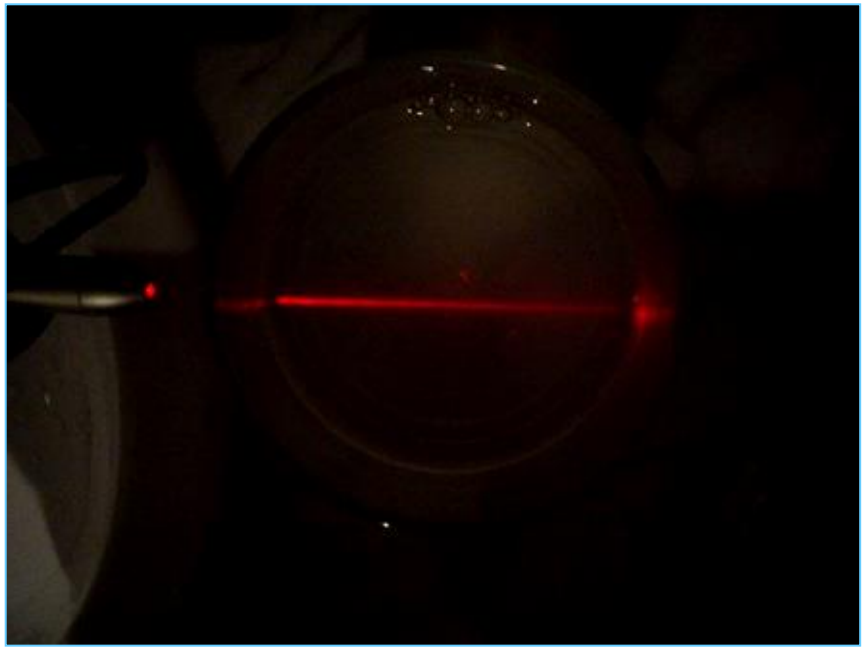




High Tech Scattering Experiment II

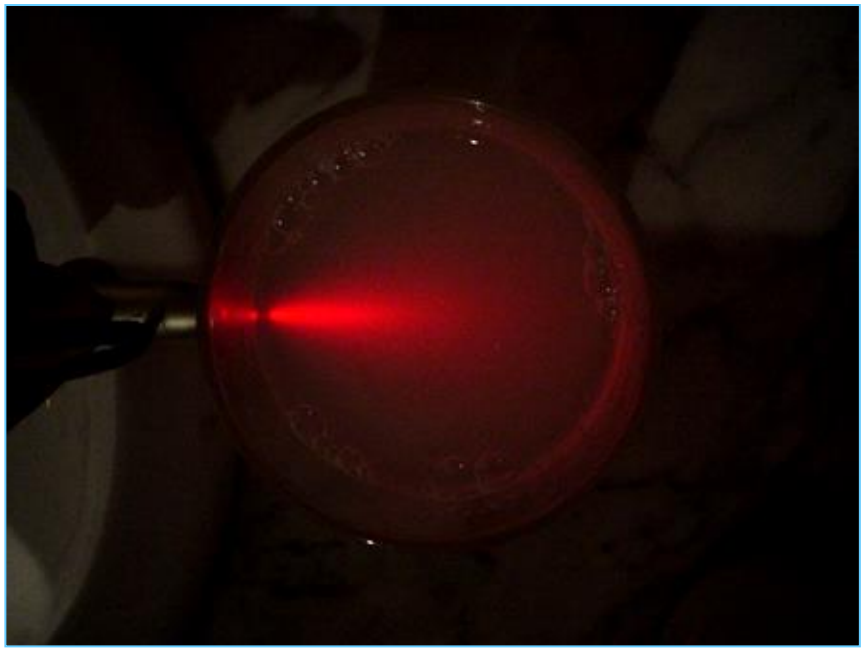


Volume Scattering



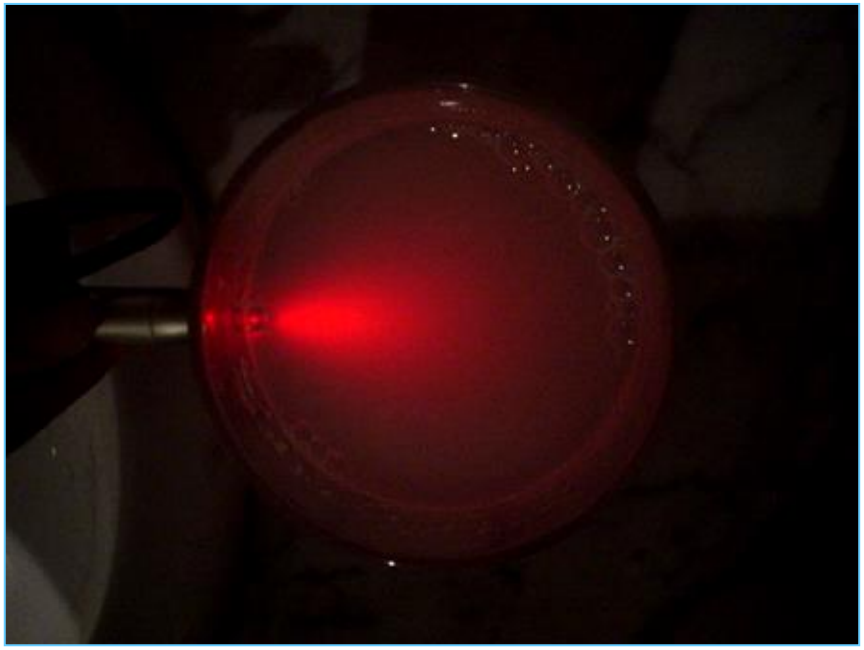
1 Drop

Volume Scattering



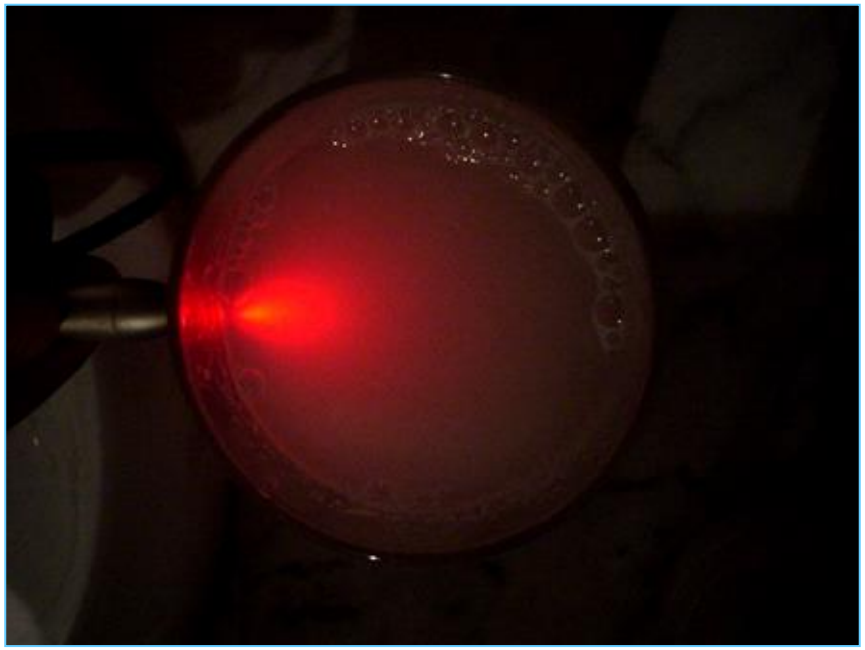
4 Drops

Volume Scattering



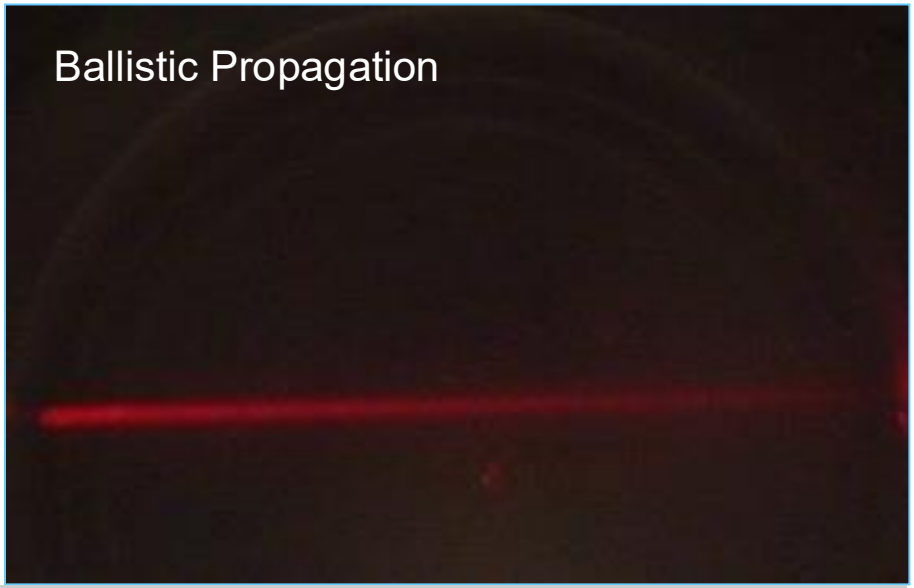
8 Drops

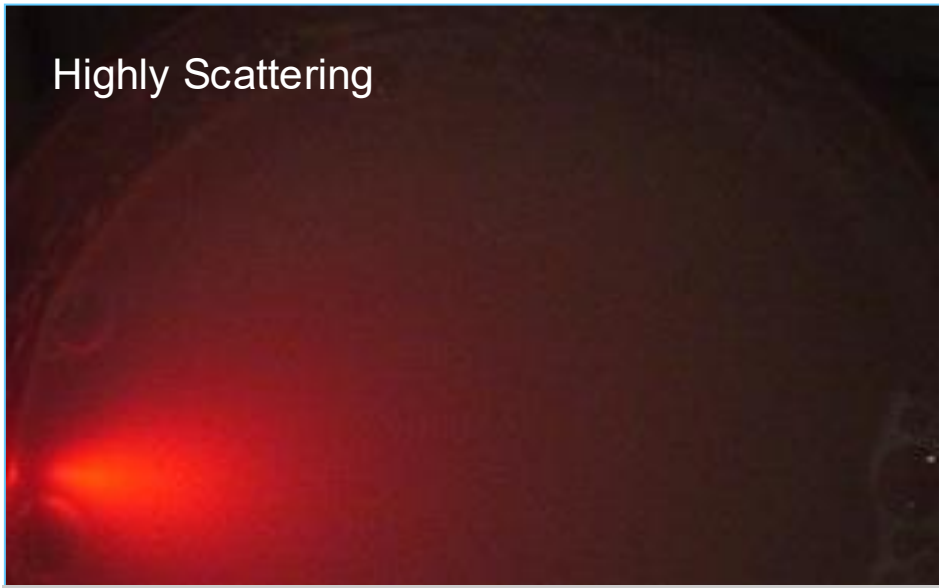
Volume Scattering

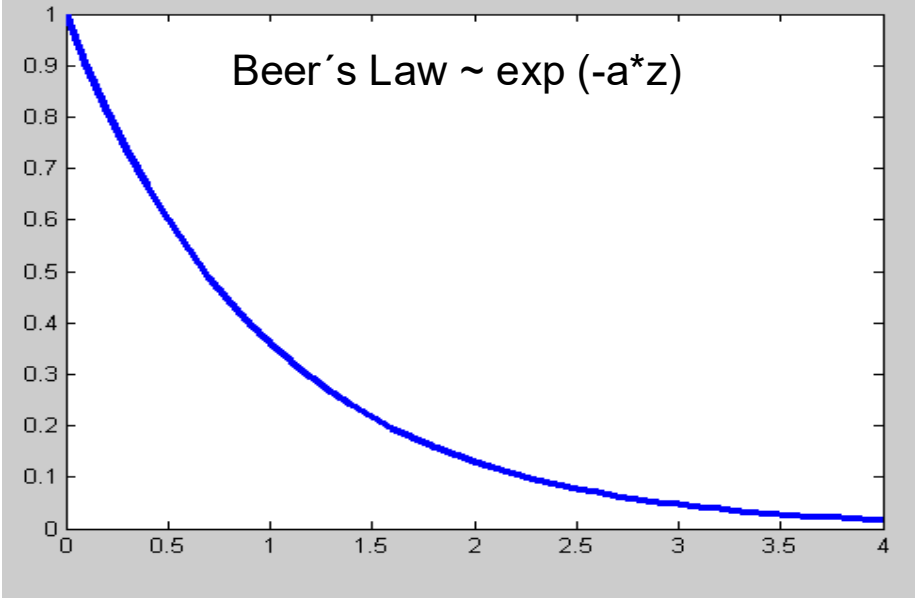


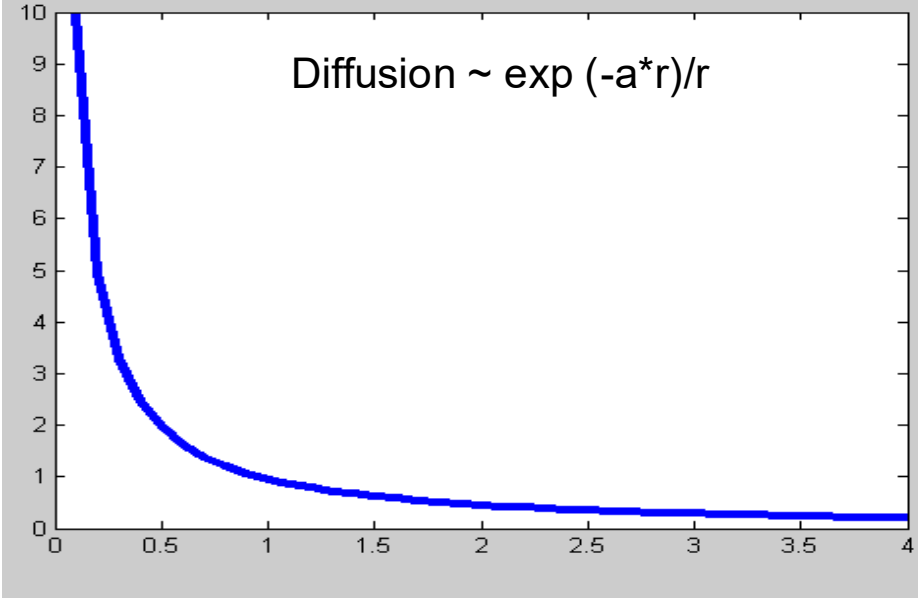
12 Drops

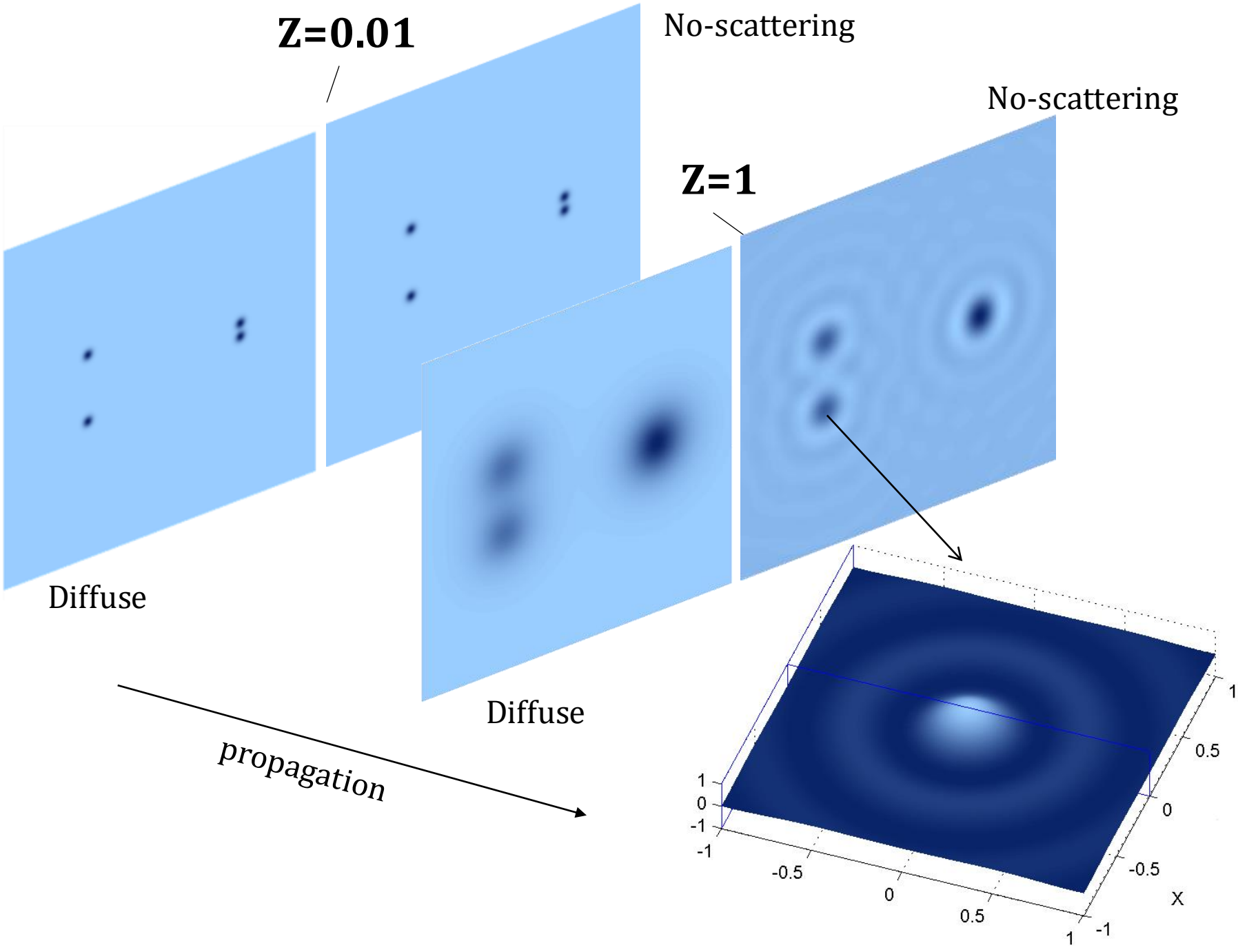
Scattering

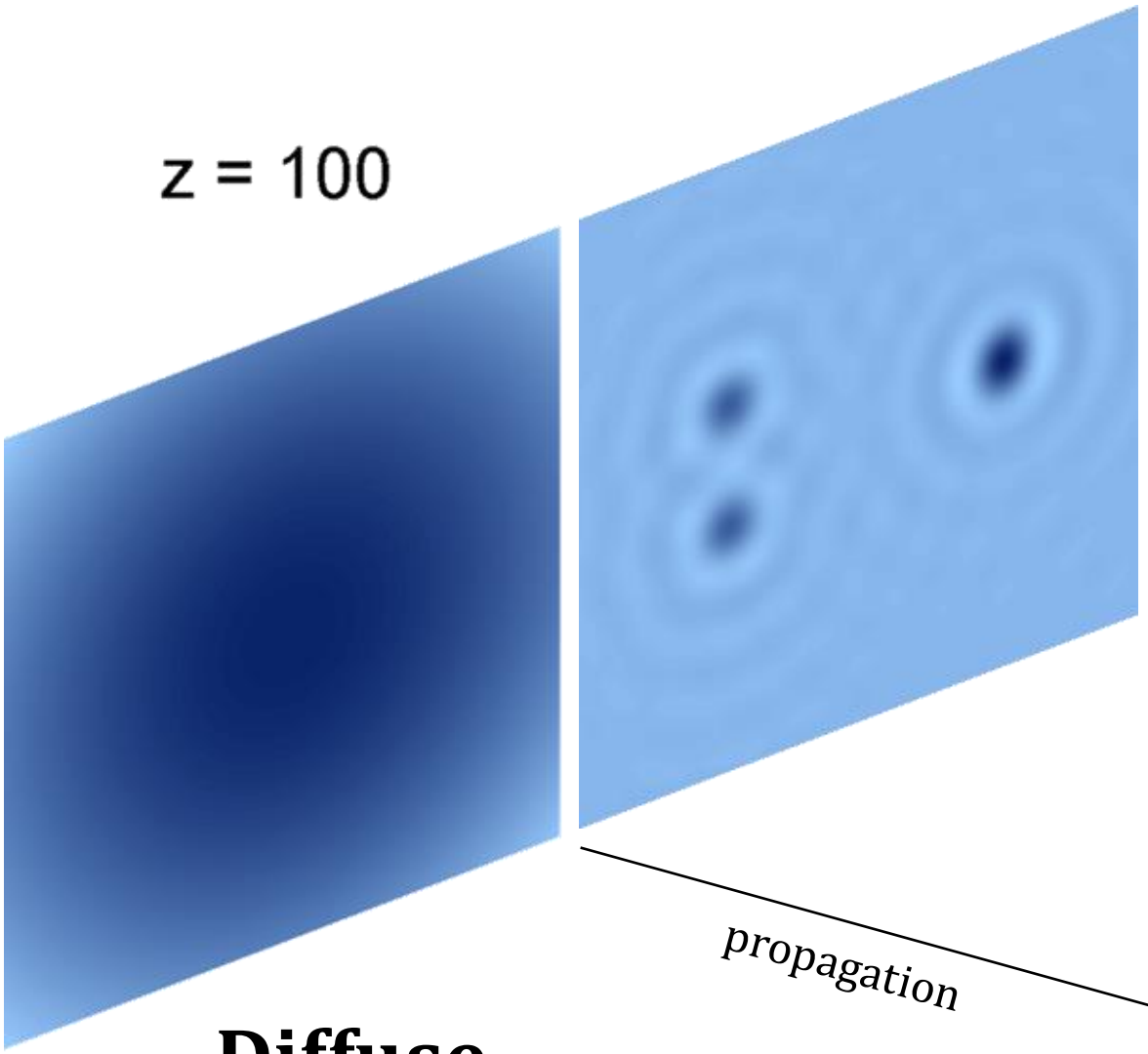

Ballistic Propagation


Highly Scattering


Beer´s Law ~ exp (-a*z)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
0.5
1
1.5
2
2.5
3
3.5
4


Diffusion ~ exp (-a*r)/r
10
9
8
7
6
5
4
3
2
1
0
0
0.5
1
1.5
2
2.5
3
3.5
4

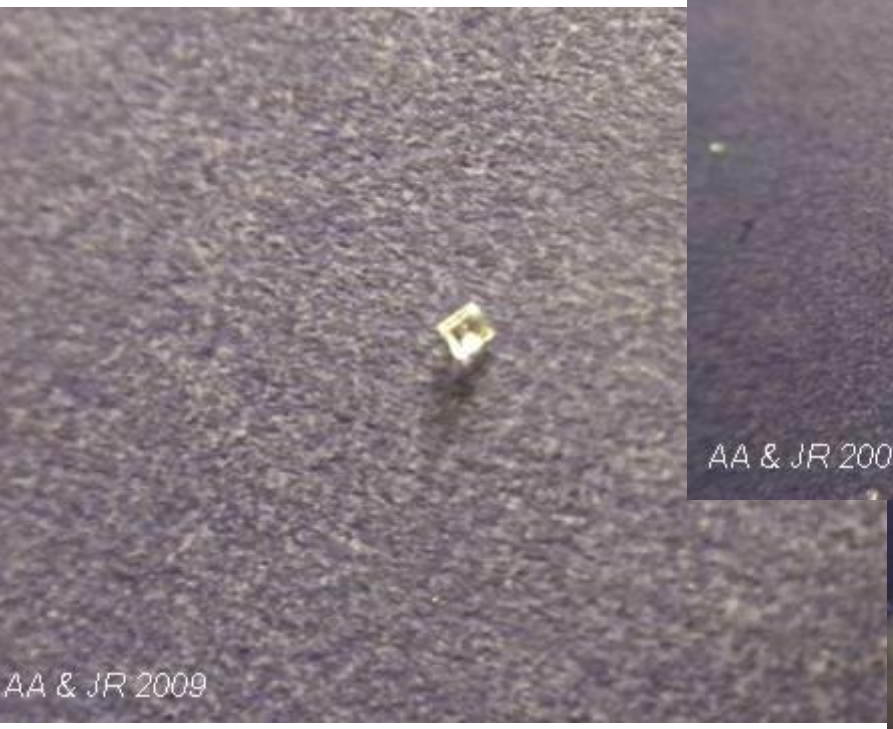

Z=0.01
No-scattering
No-scattering
Z=1
Diffuse
Diffuse
propagation
1
0
-1
-1
-0.5
0
0.5
1
1
0.5
0
-0.5
-1
X



No-scattering

Diffuse

Other stuff that multiply scatters light

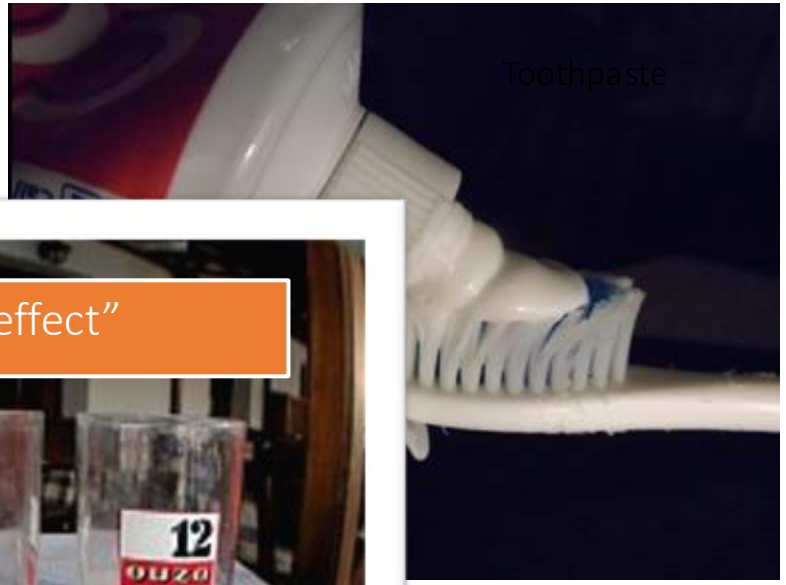




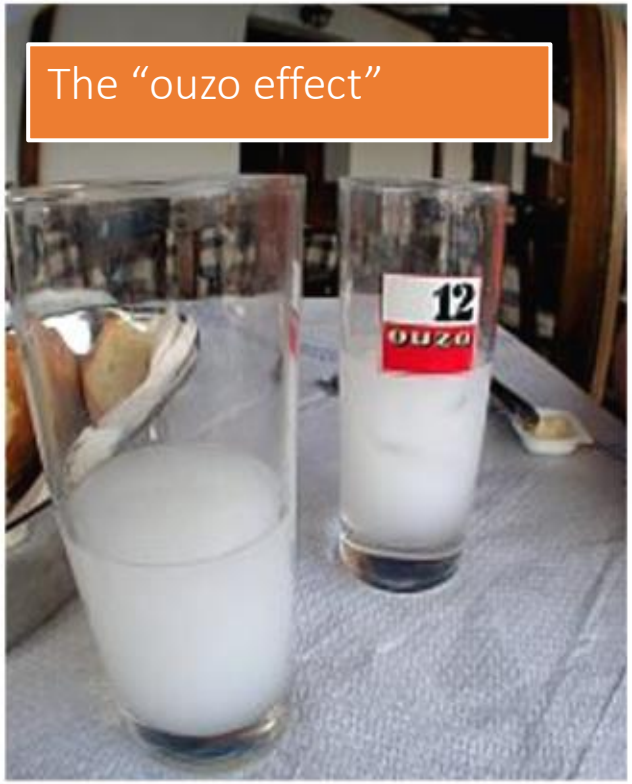


Sugar





The “ouzo effect”



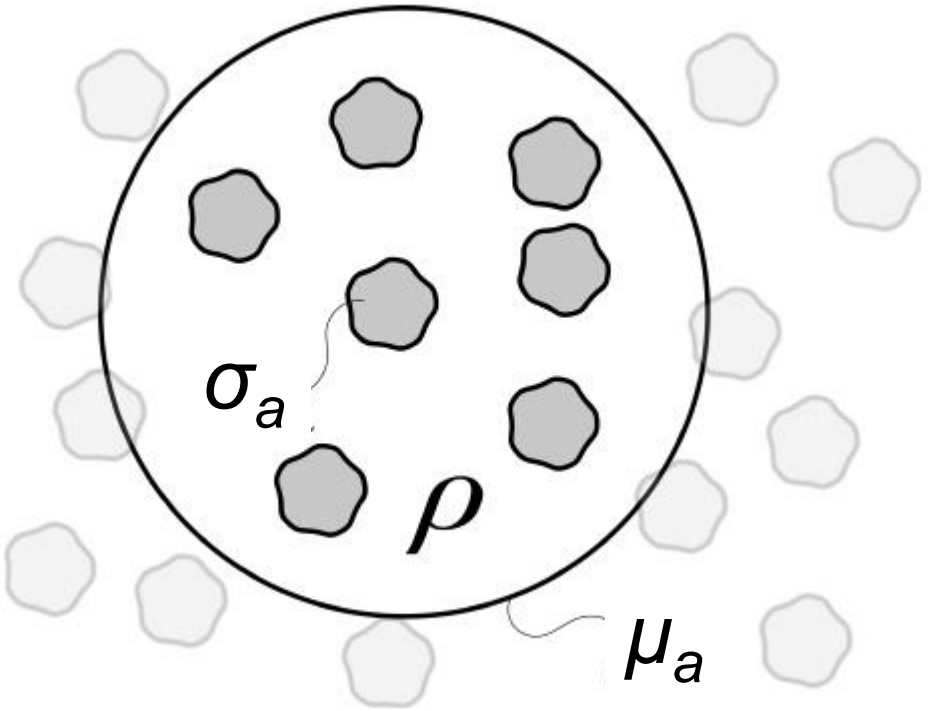


om a nice slice of jamón serrano



Absorption

Statistical Description of Optical Properties

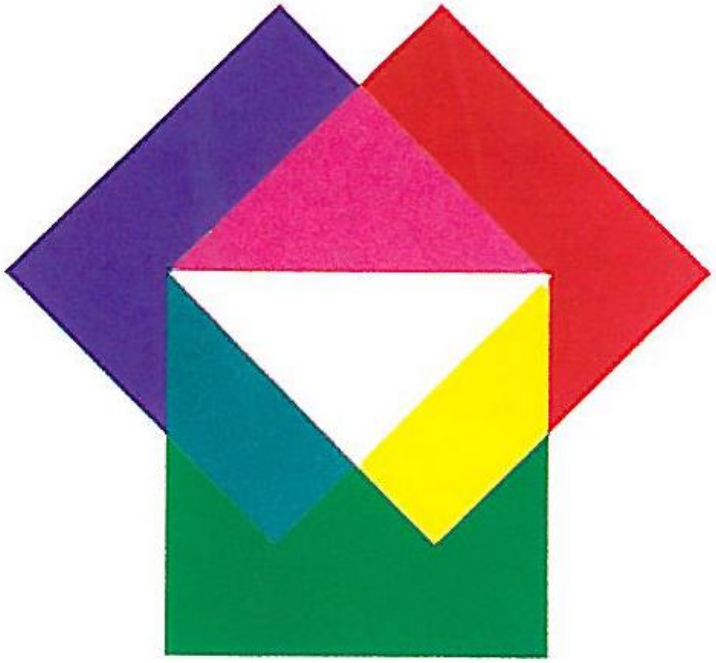


In the case where we have a collection of these particles at a certain density, the amount of scattering and absorption will depend on the density of these.

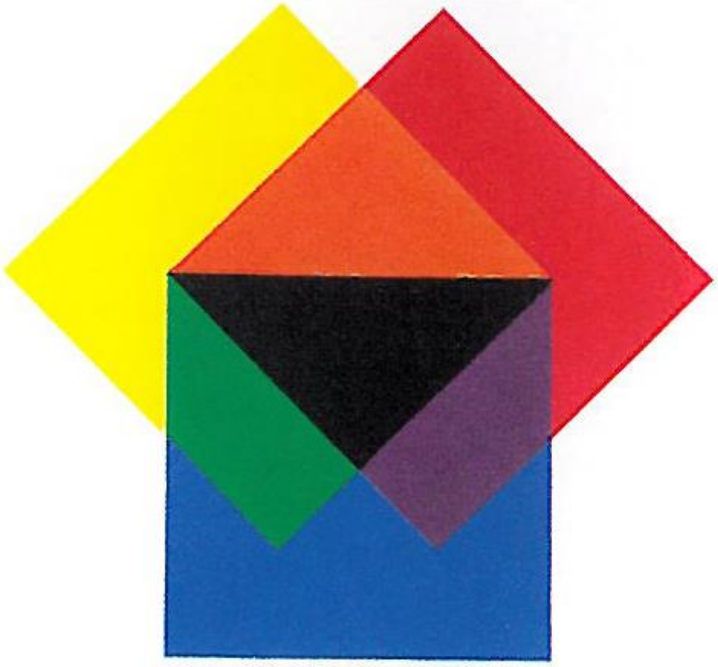
Absorption Coefficient

$$\mu_a = \rho\sigma_a. \qquad \left(cm^{-1}\right)$$

Light Absorption



Additive Primaries
(adding light)

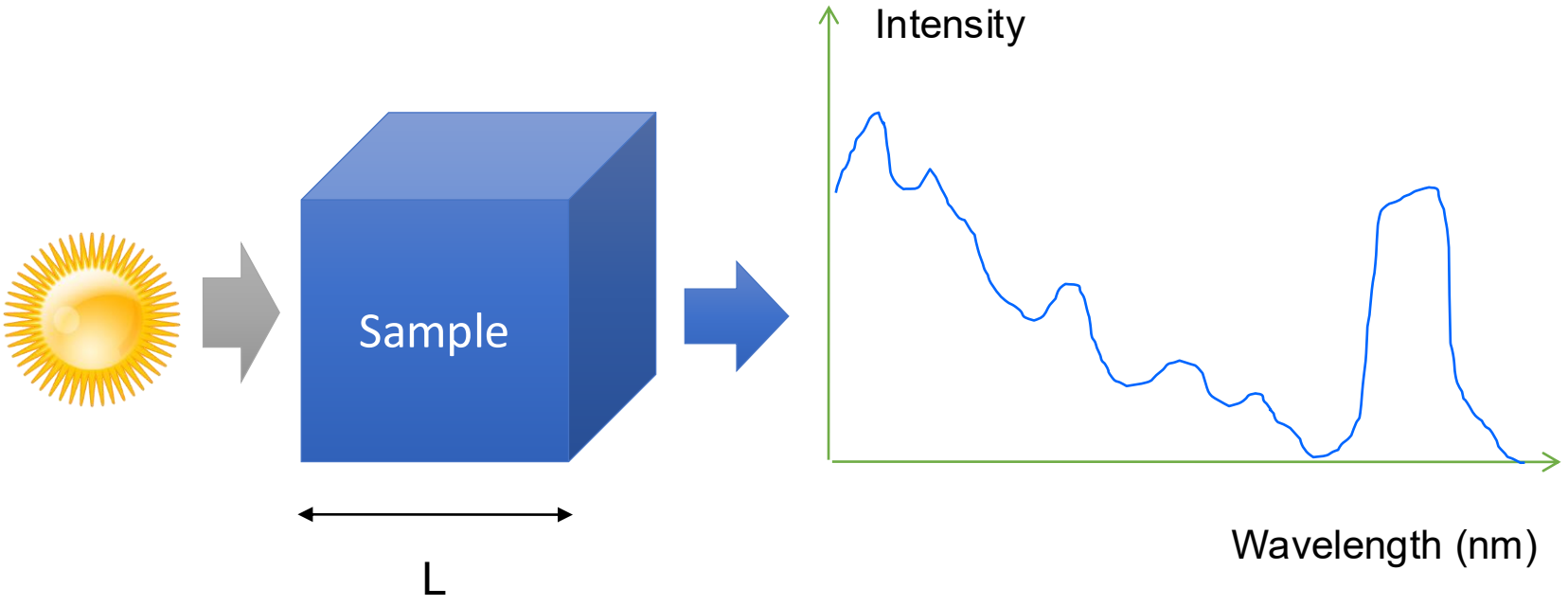


Substractive Primaries
(adding absorption)

Light Absorption

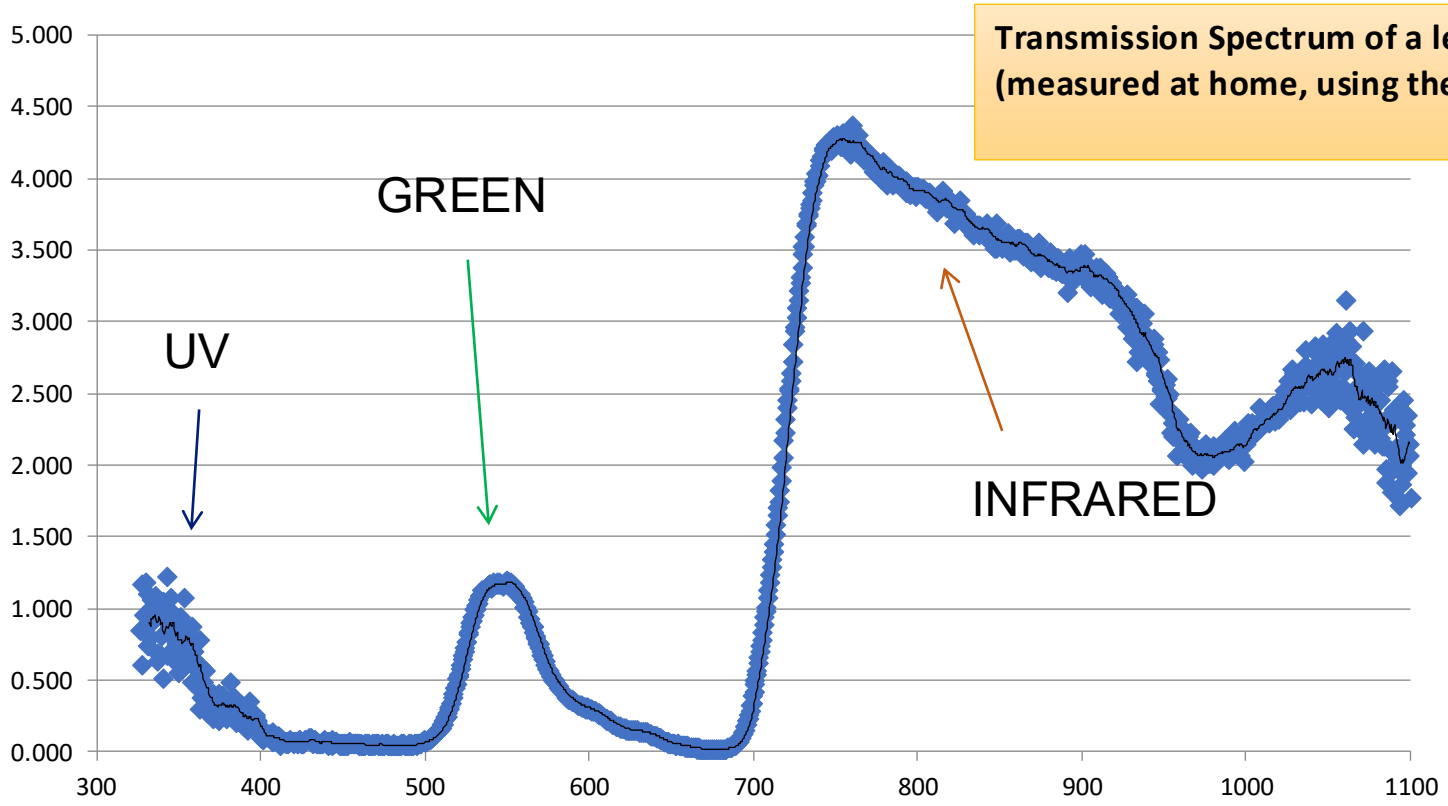
This approach is the one used in a spectrograph

$$I(L) = I_0 \exp -\mu_a L$$



Light Absorption

Which, seen through our spectrograph will result in an **transmission curve**, i.e. a curve that represents intensity versus wavelength (i.e. the amount of energy present for each frequency measured). When compared to the incident light, we will have the **Absorption curve**.



This is a direct measurement of the energy levels present in this particular leaf.

Tissue Absorption

- Main Absorbers in Tissue:
  - Blood
  - Water
  - Skin (melanin)

Blood Absorption

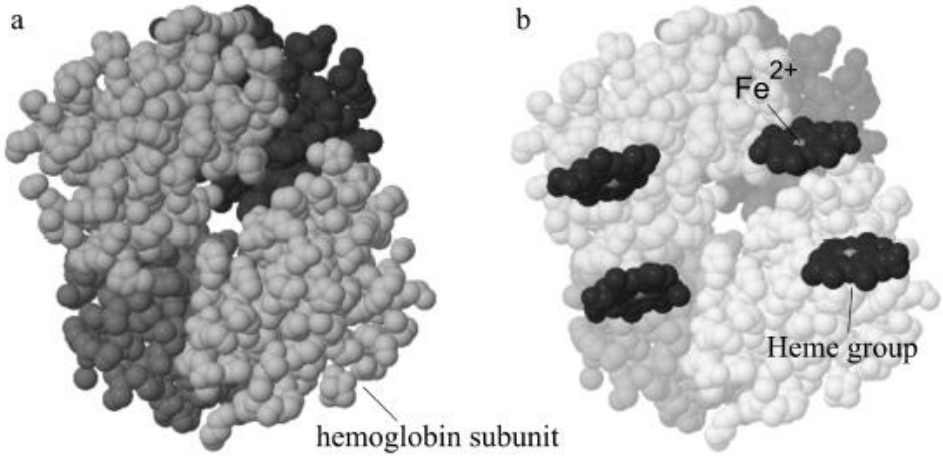
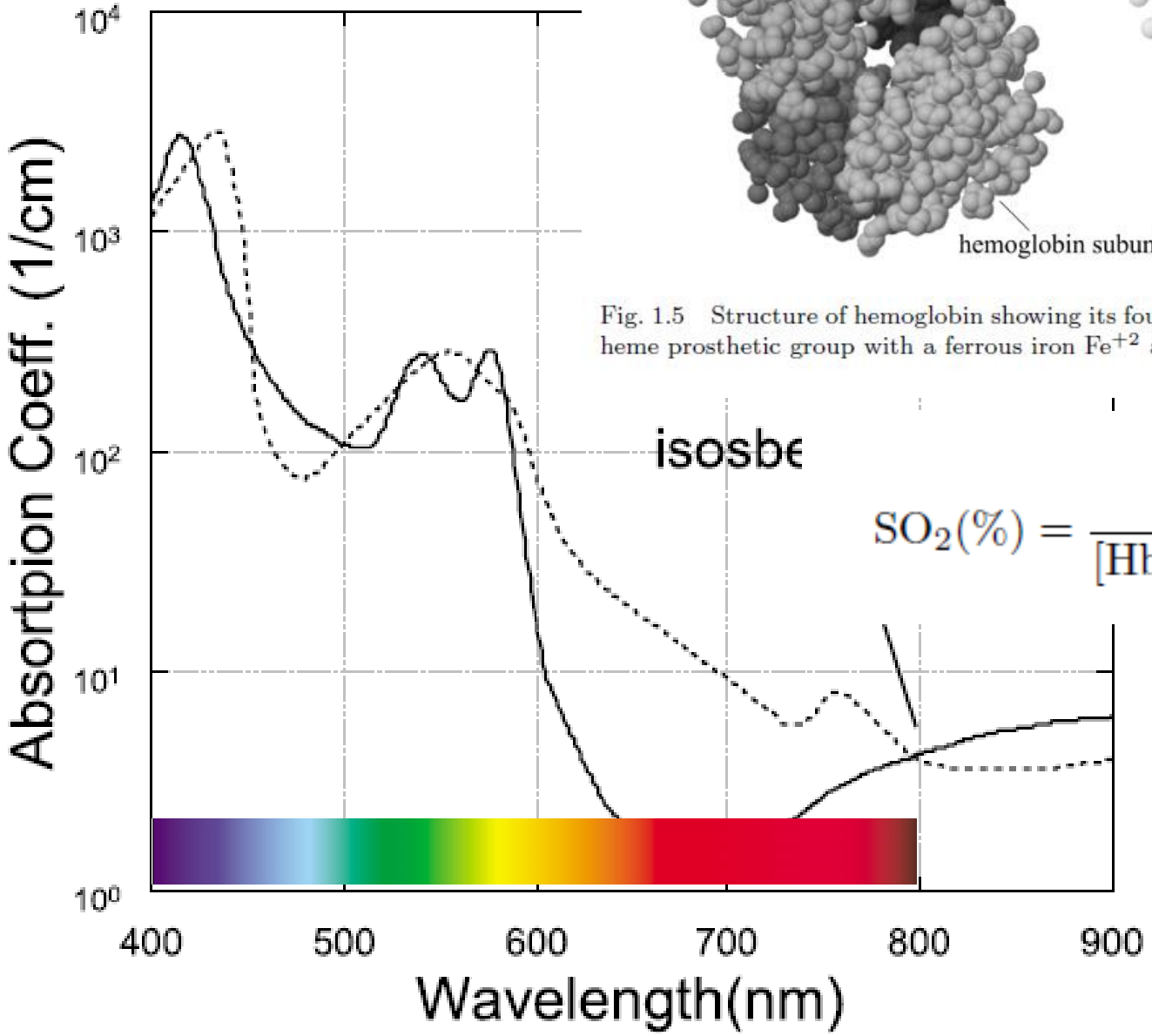


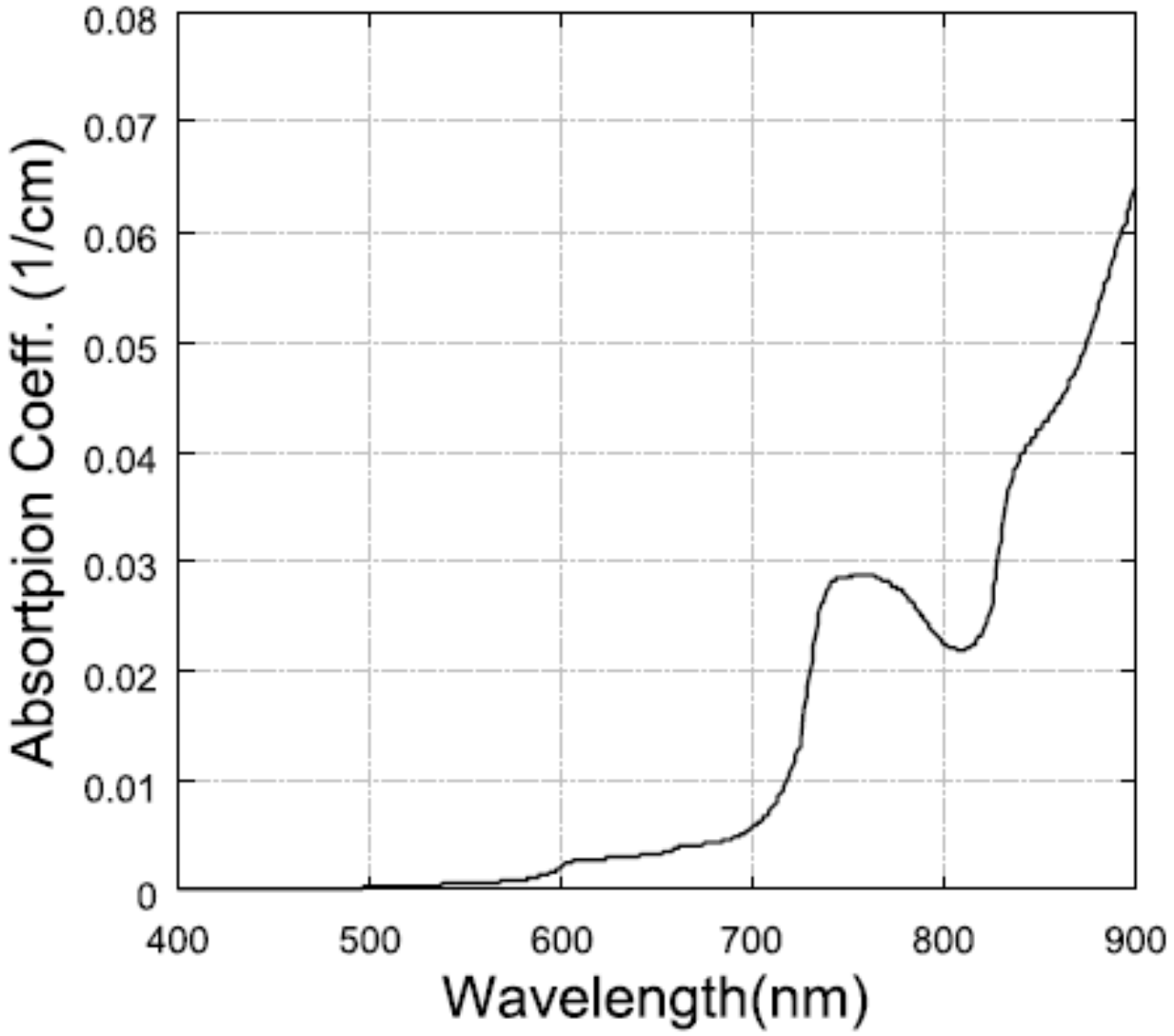
Fig. 1.5 Structure of hemoglobin showing its four subunits (a) with their corresponding heme prosthetic group with a ferrous iron Fe^{+2} atom (b).



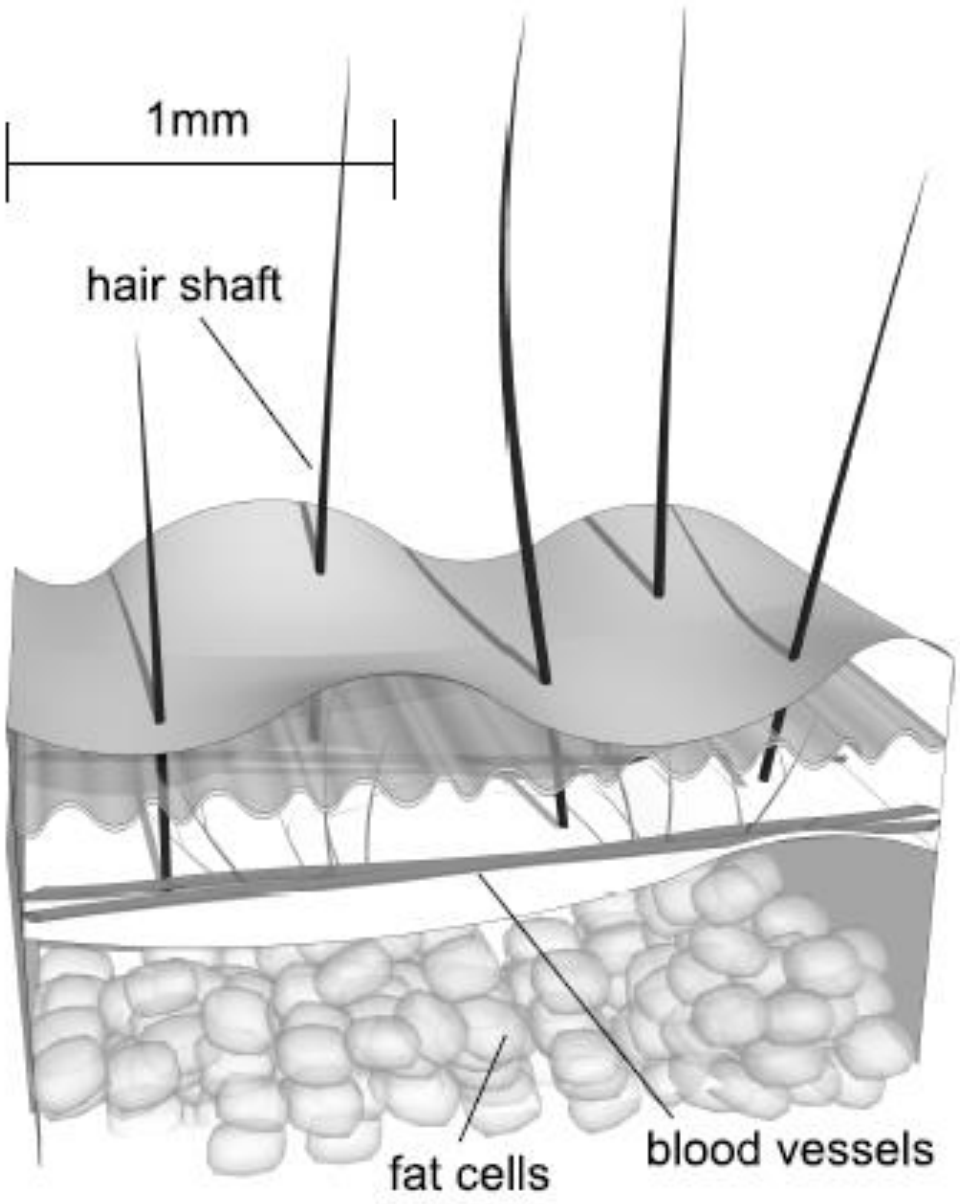
isosbe

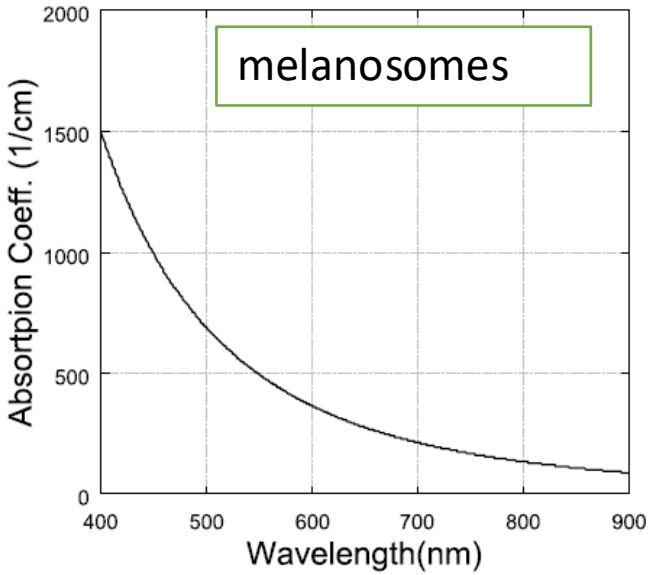
$$SO_2(\%) = \frac{[HbO_2]}{[HbO_2] + [Hb]} \times 100,$$

Absorption Water

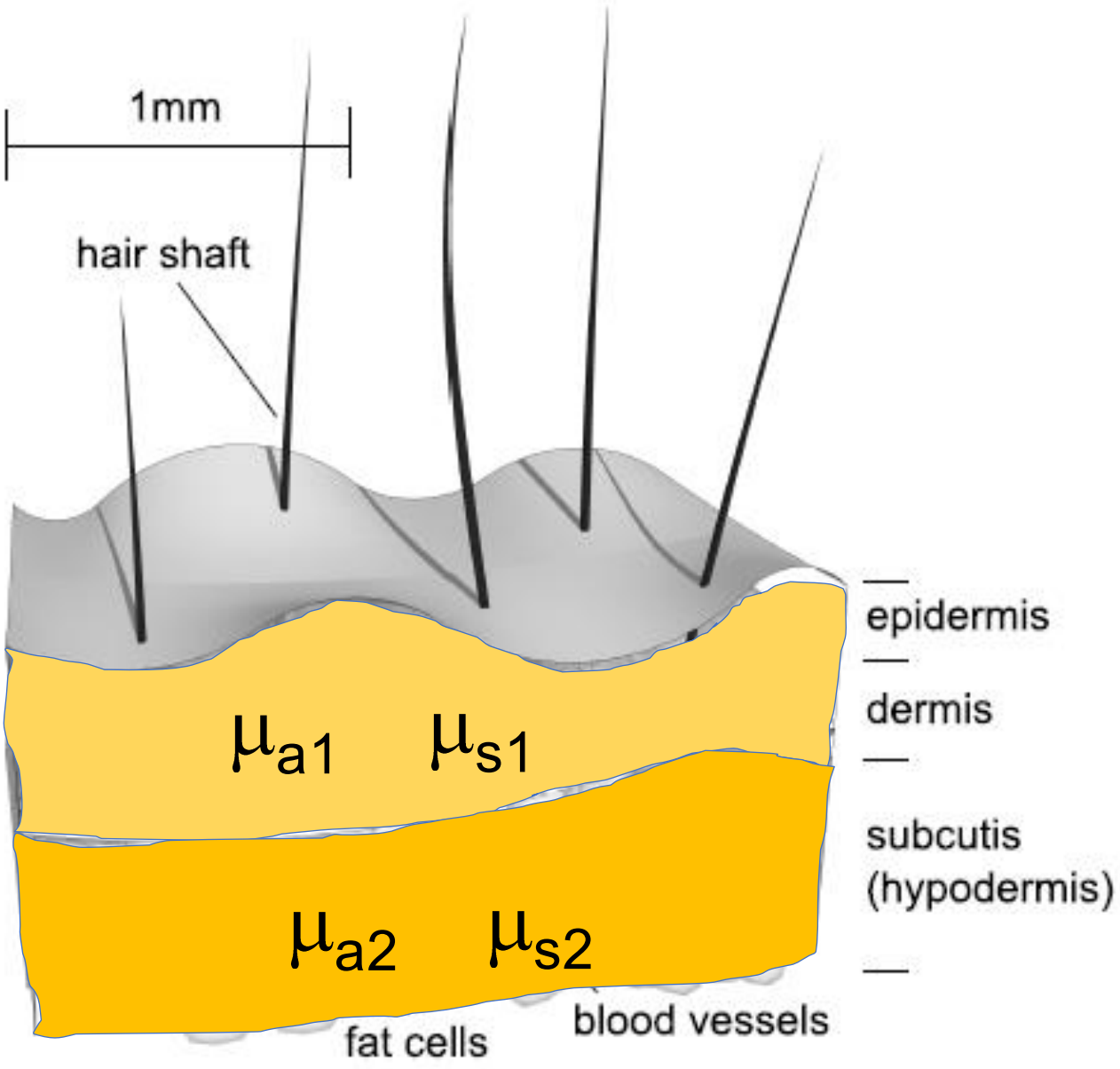


Effect of Skin

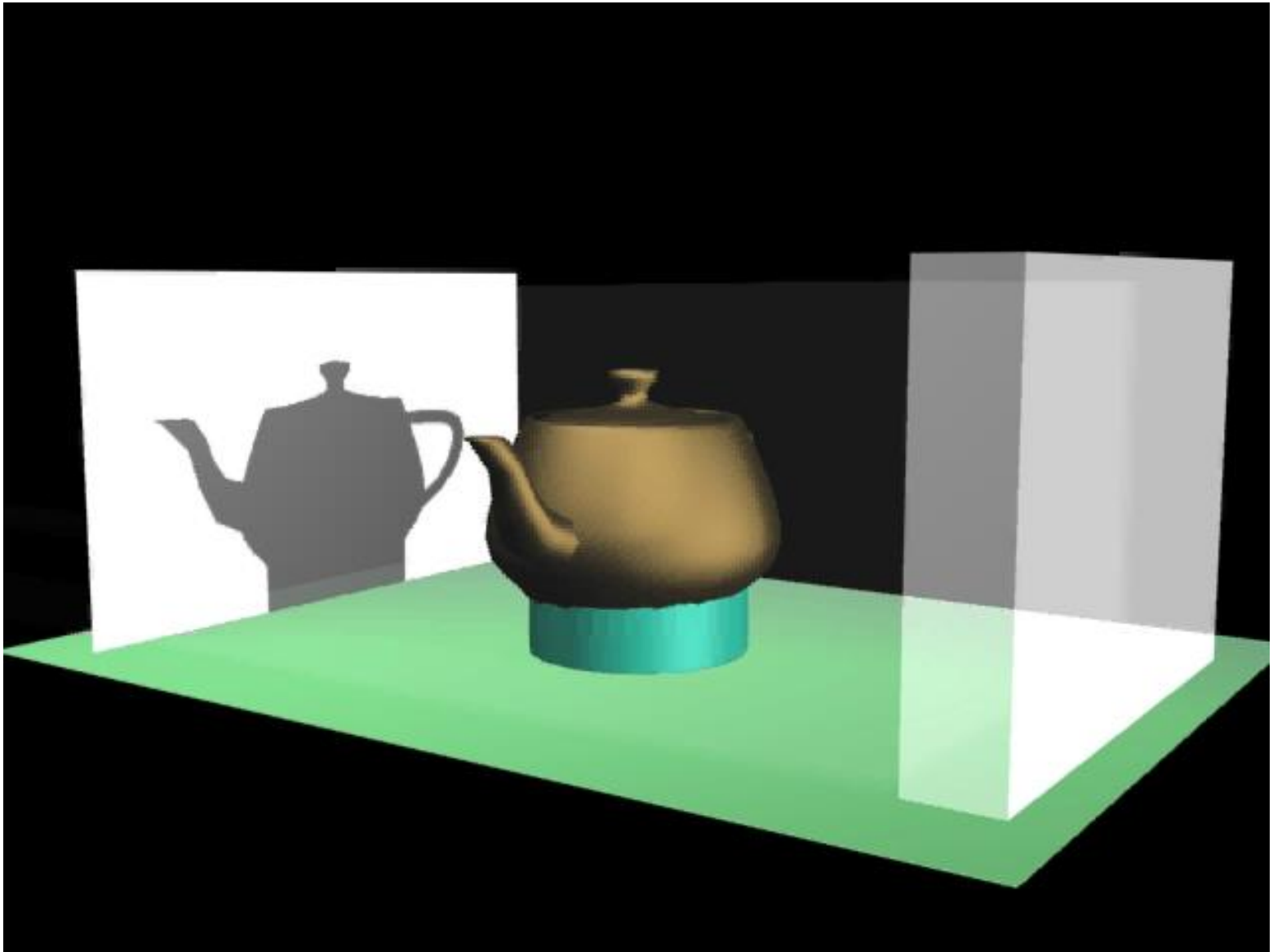




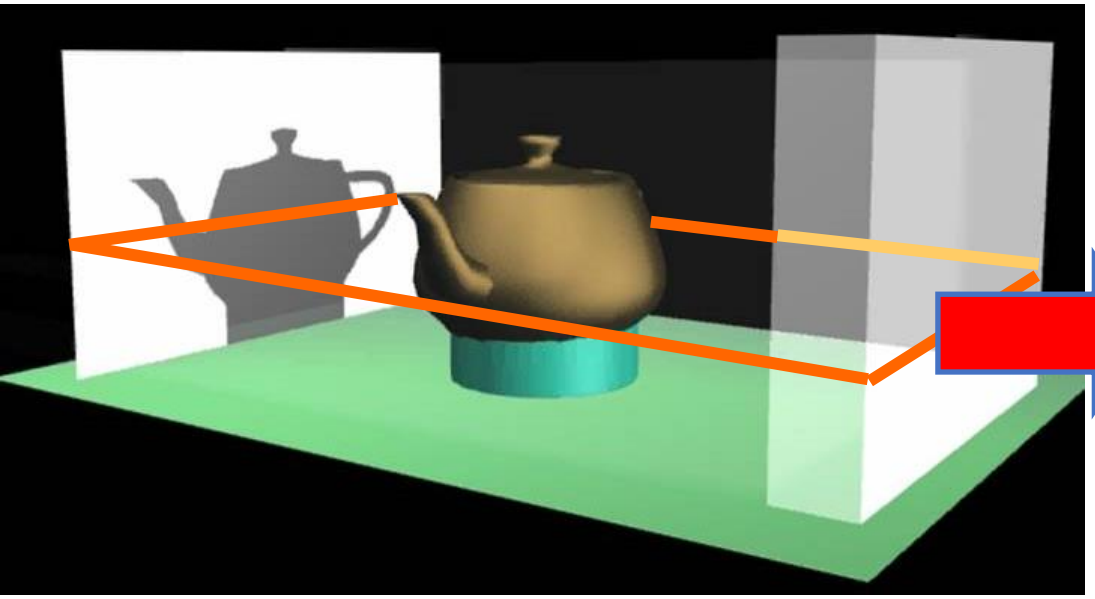
Statistical Description of optical properties

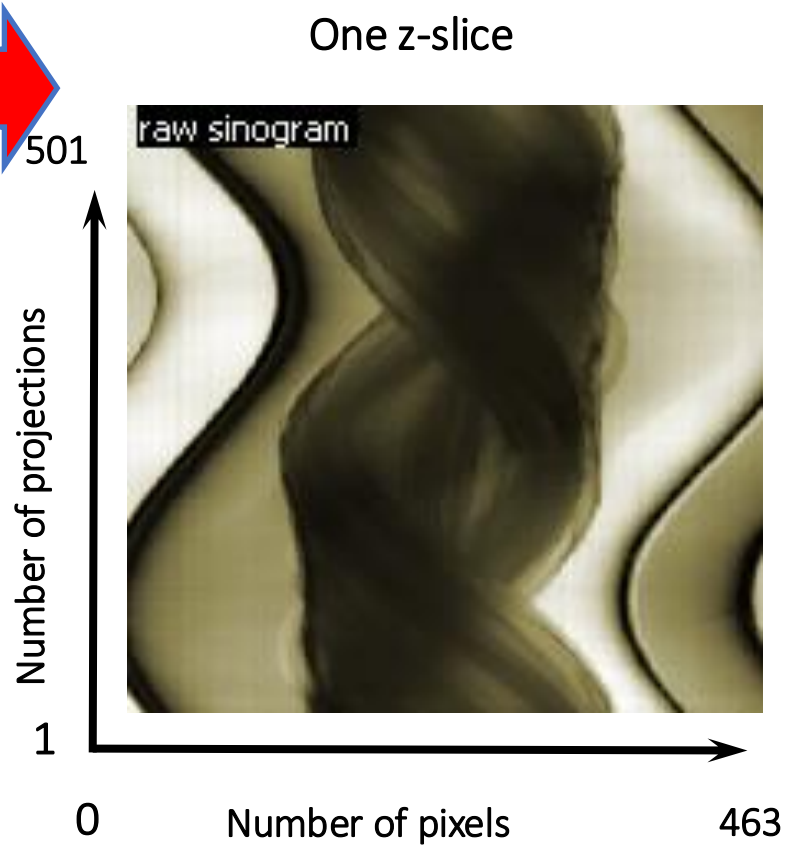


Radon Transform



Radon Transform



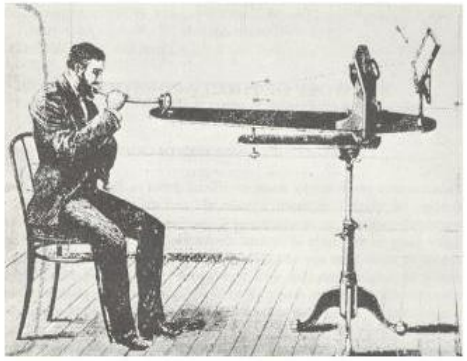


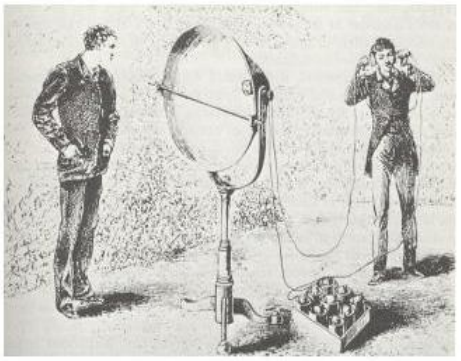
Photoacoustics

Photoacoustic imaging generates contrast using light, and measures the acoustic wave generated

Photoacoustic Effect

The basis of the Photoacoustic (PA) effect is the non-radiative release of part of the light absorbed by the molecules in a given material, resulting in local heating and thus the generation of a pressure (or acoustic) wave propagating away from the heat source.
Since the discovery of the PA effect, in 1880 by **Alexander Graham Bell**, techniques related to its application have evolved to become very efficient methods for thermal and optical characterization of light absorbing materials.









Bell, A. G. *Philos. Mag.* (*Fifth Series*) **1881**, *11*, 510-528.

Photoacoustic Effect

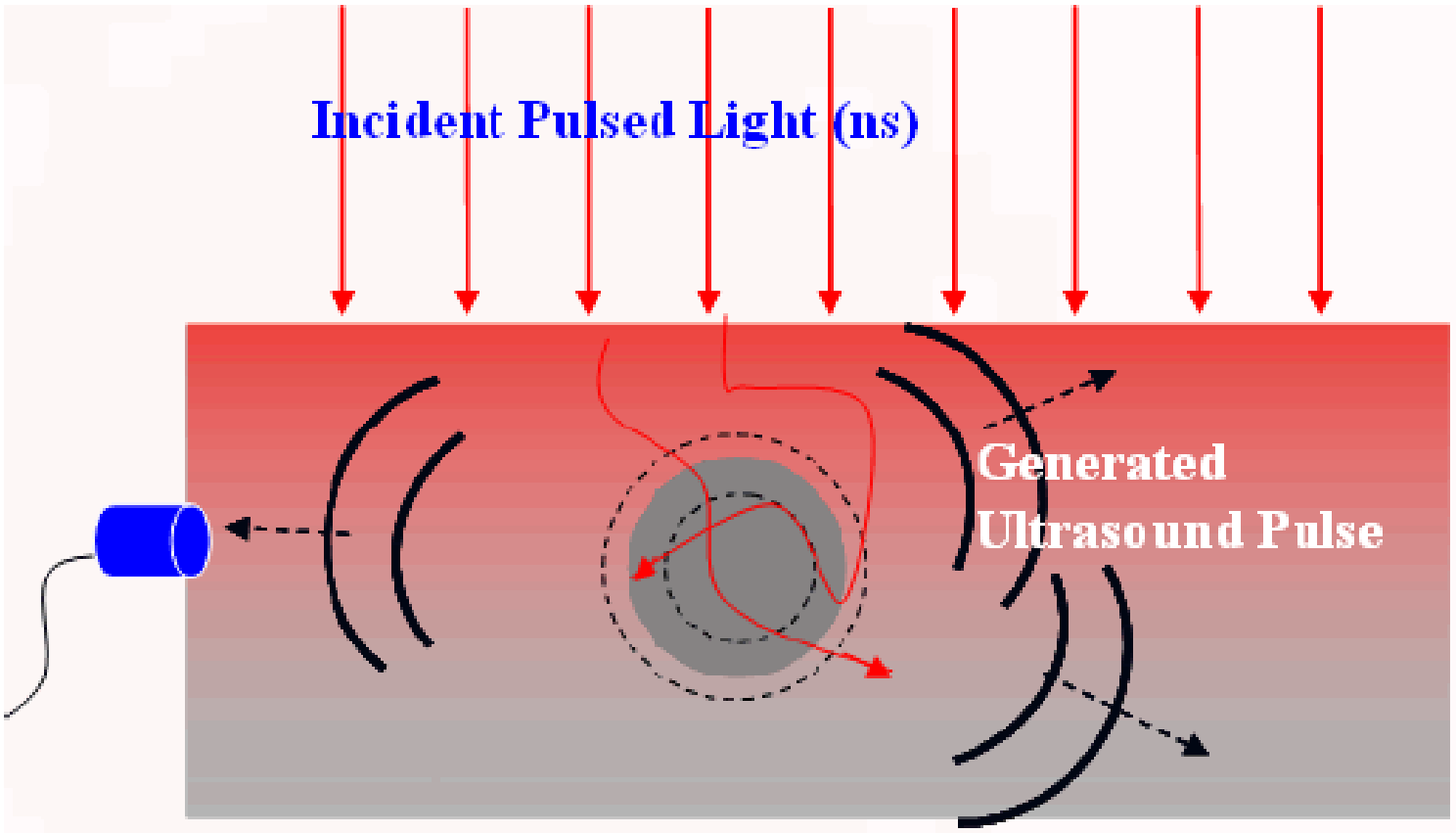


Figure 1: Schematic of photoacoustic effect.

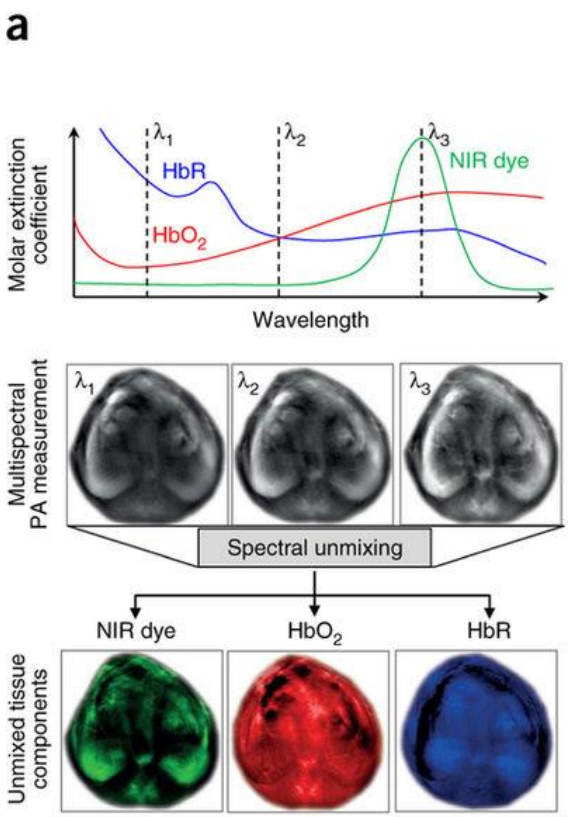
Multiscale PAT of single cells, whole-body small animals, and humans.

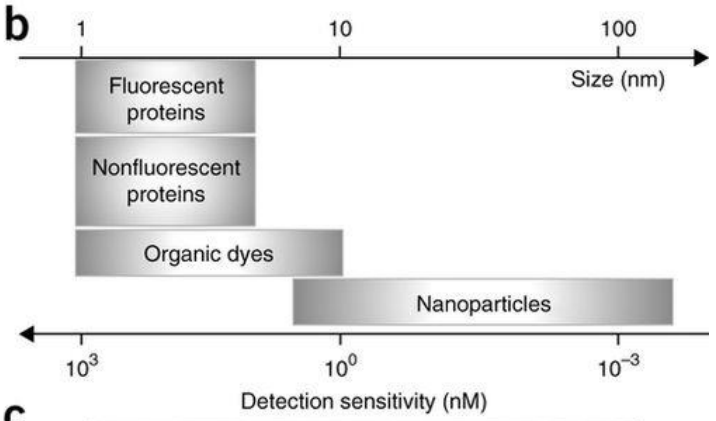
Label-free photoacoustic imaging of human palmar vessels: a structural morphological analysis. Y. Matsumoto et al (2018), Scientific Reports https://doi.org/10.1038/s41598-018-19...

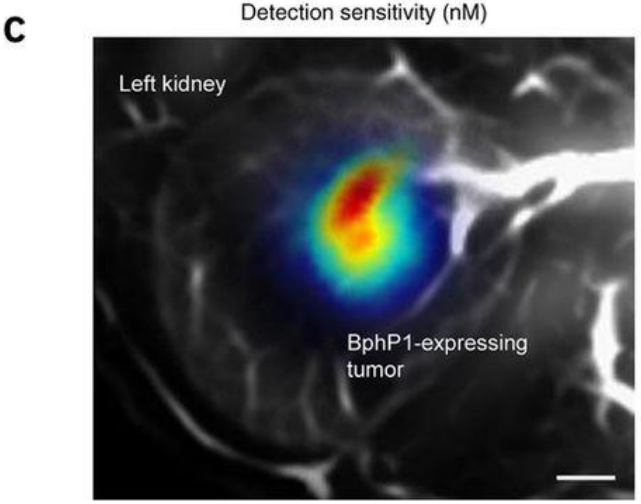


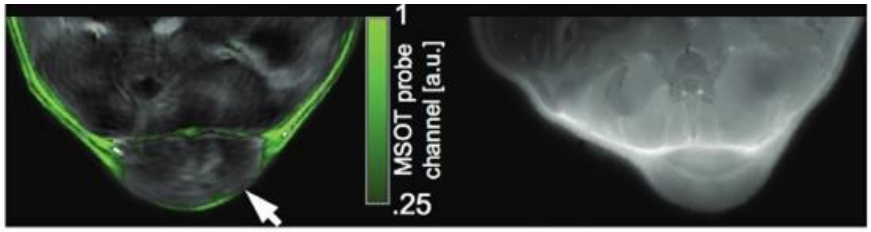
https://www.youtube.com/watch?v=_Y4O8loaylY

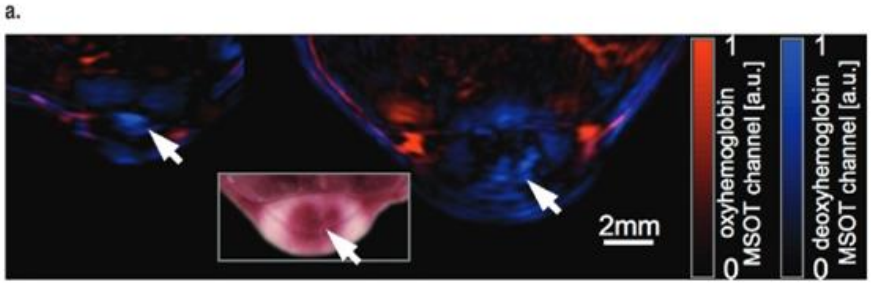
Multispectral











V. Ntizachristos, TUM

Scale bar, 1 mm.

Neuroimaging

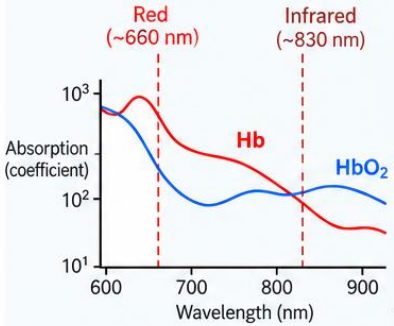
Both fDOT and fMRI measure brain activity through changes in oxygenation

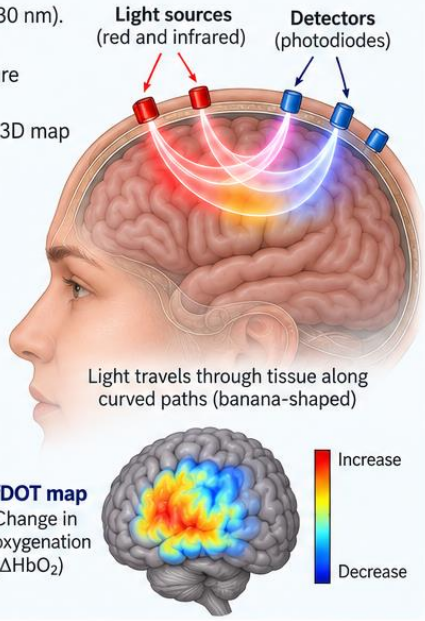
Neuronal activity increases local blood flow, changing the relative amounts of oxyhemoglobin (HbO_2) and deoxyhemoglobin (Hb). fDOT and fMRI detect these changes using different physical mechanisms.

Functional Diffuse Optical Tomography (fDOT)

Optical measurement of hemoglobin oxygenation

- Uses near-infrared light (~660 and ~830 nm).
- HbO_2 and Hb absorb light differently.
- Multiple sources and detectors measure light at many locations.
- A reconstruction algorithm provides a 3D map of changes in [HbO_2] and [Hb].





Functional MRI (fMRI)

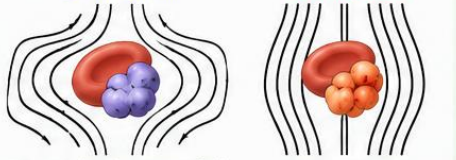
Magnetic measurement of hemoglobin oxygenation (BOLD)

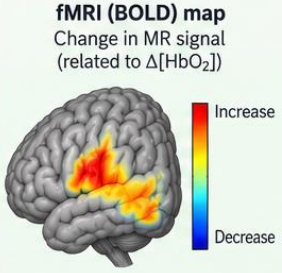
- Uses a strong magnetic field (MRI scanner).
- Deoxyhemoglobin (Hb) is paramagnetic, while oxyhemoglobin (HbO_2) is diamagnetic.
- Changes in the relative amount of Hb vs. HbO_2 alter the local magnetic field and the MR signal (BOLD contrast).
- Provides a 3D map of brain activity (indirectly related to changes in oxygenation and blood flow).



MRI scanner

Effect of oxygenation on MR signal





Key point

Both fDOT and fMRI measure brain function by detecting changes in oxygenation:
fDOT uses the different light absorption of HbO_2 and Hb, while fMRI uses the different magnetic properties (paramagnetism) of Hb.

Biomedical analysis technologies
Optical analysis systems in laboratories and hospitals

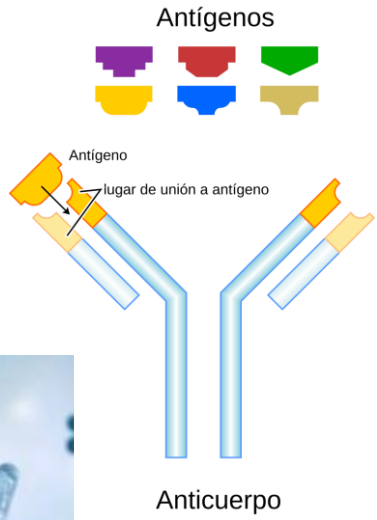
Introduction to Bioengineering

Jorge Ripoll
Jorge.Ripoll@uc3m.es

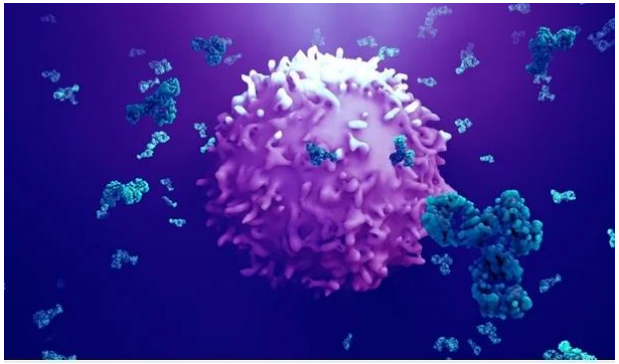
ELISA

Enzyme- Linked immunosorbent assay

Enzyme- linked immunosorbent assay
Enzyme-linked immunosorbent assay



Antigen



Antibody

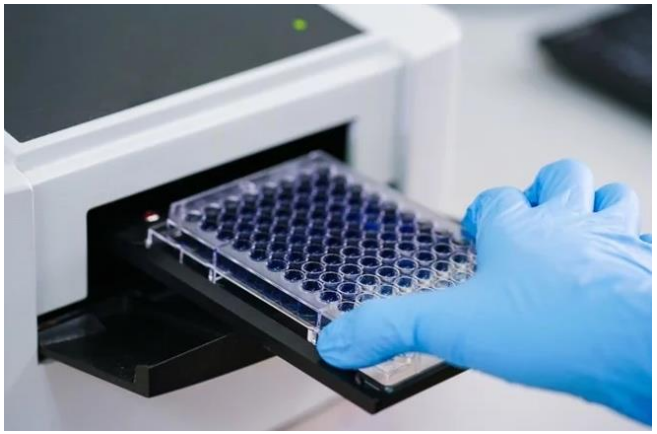


Trigger an immune response

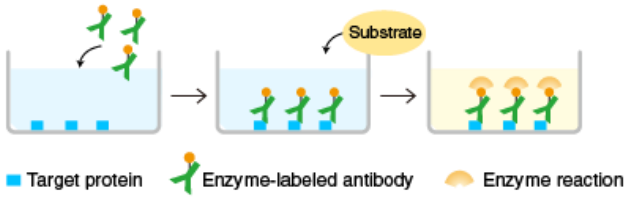
Immune-system response to identify and neutralize antigens

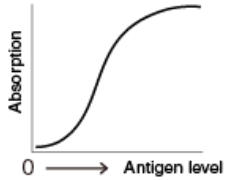
ELISA

Enzyme- Linked immunosorbent assay

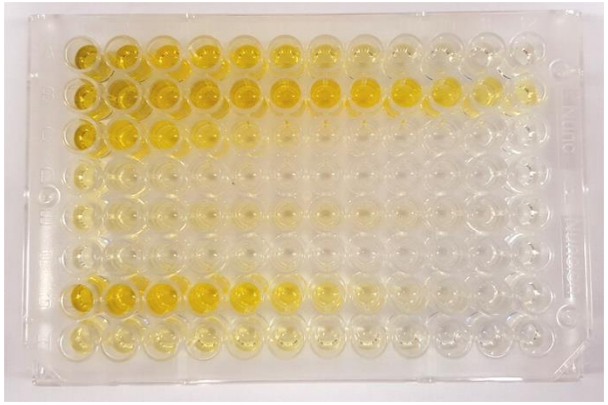


Direct ELISA





Standard calibration curve – Color intensity is used to quantify the antigen in the sample.



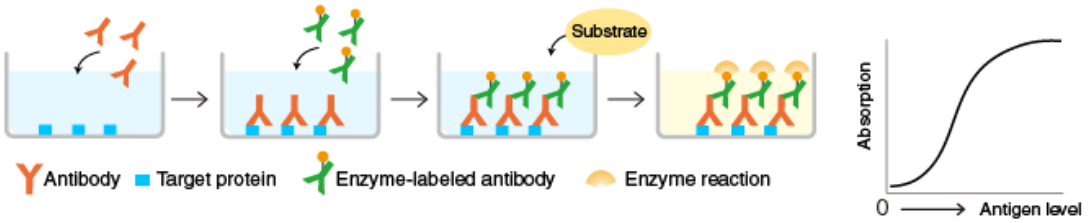
The simplest format
Enzyme directly conjugated to the antibody

1. **Antigen immobilization. The antigen (the molecule of interest, such as a protein) adheres directly to the surface.**
2. **Blocking. A blocking agent (e.g., bovine serum albumin) is added to prevent nonspecific binding.**
3. **Addition of the labeled antibody. An antigen-specific antibody conjugated to an enzyme (e.g., horseradish peroxidase, HRP, or alkaline phosphatase) is added.**
4. **Addition of the substrate. A chromogenic substrate is added, which the antibody-linked enzyme converts into a colored product.**

ELISA

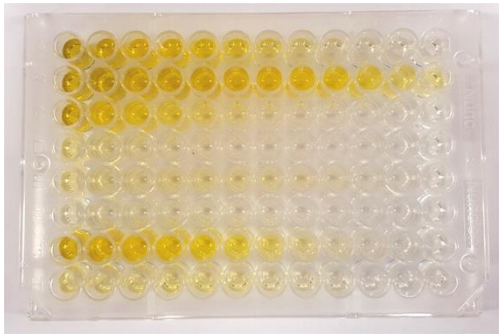
Enzyme- Linked immunosorbent assay

Indirect ELISA



Primary antibody – Specifically recognizes the antigen.
Secondary antibody – Carries the enzyme.
Recognizes and binds to the primary antibody (NOT the antigen).

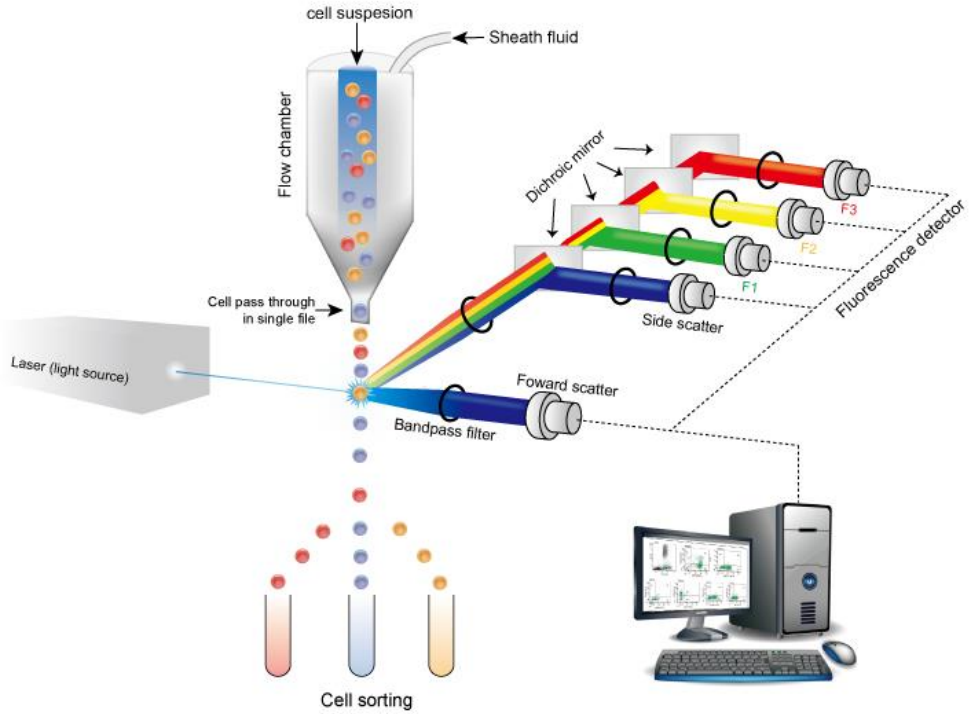
More sensitive than the direct method



1. **Antigen immobilization**
2. **Blocking**
3. **Addition of the primary antibody. An antigen-specific antibody without an enzyme label is added. It binds directly to the immobilized antigen.**
4. **Wash. Unbound primary antibodies are removed.**
5. **Addition of the secondary antibody. A secondary antibody that recognizes the primary antibody (e.g., anti-IgG) is added. It is enzyme-conjugated (HRP, alkaline phosphatase, etc.).**
6. **Wash. Unbound secondary antibodies are removed.**
7. **Addition of the substrate. The enzyme converts the chromogenic substrate into a colored product. Color intensity is proportional to the amount of primary antibody bound to the antigen.**

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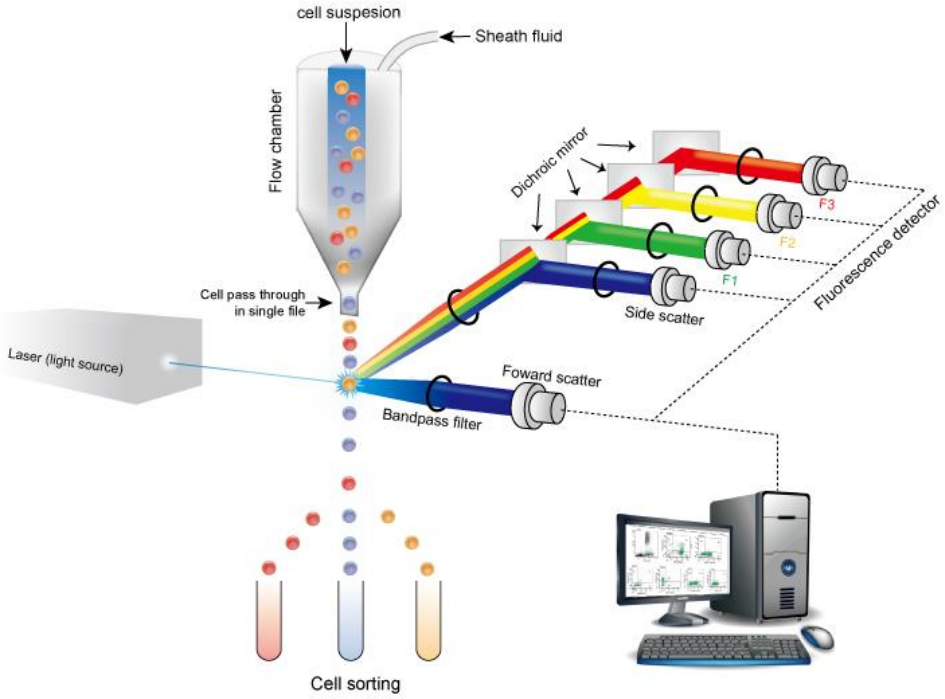
Fluorescence Activated Cell Sorting



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Fluorescence Activated Cell Sorting

Fluorescence-activated cell sorting

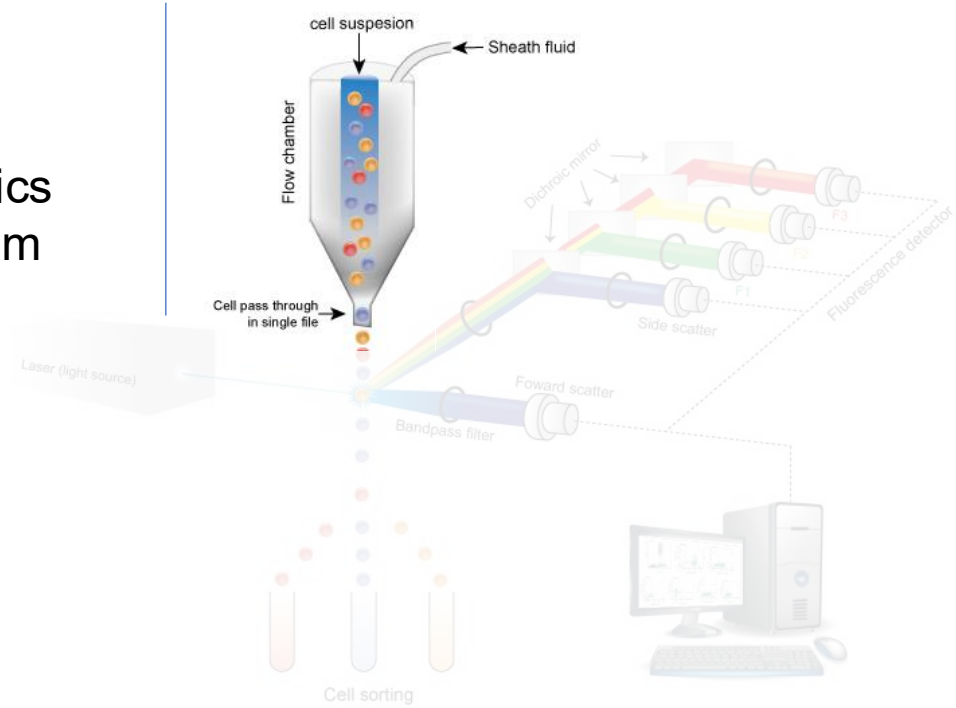


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Fluorescence Activated Cell Sorting

Fluorescence-activated cell sorting

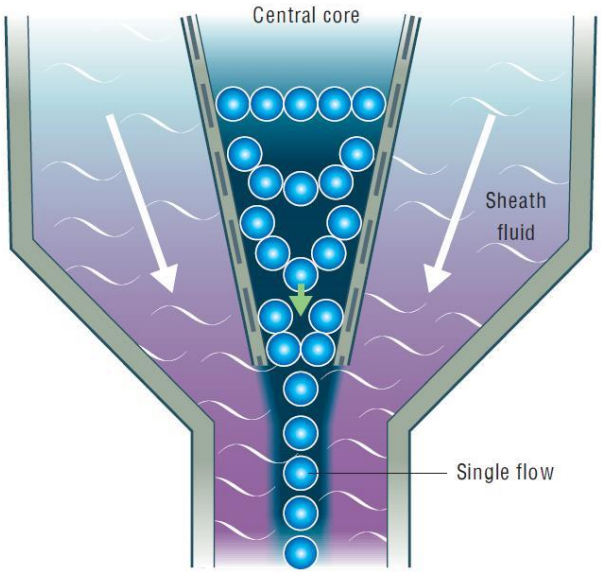
Fluidics system

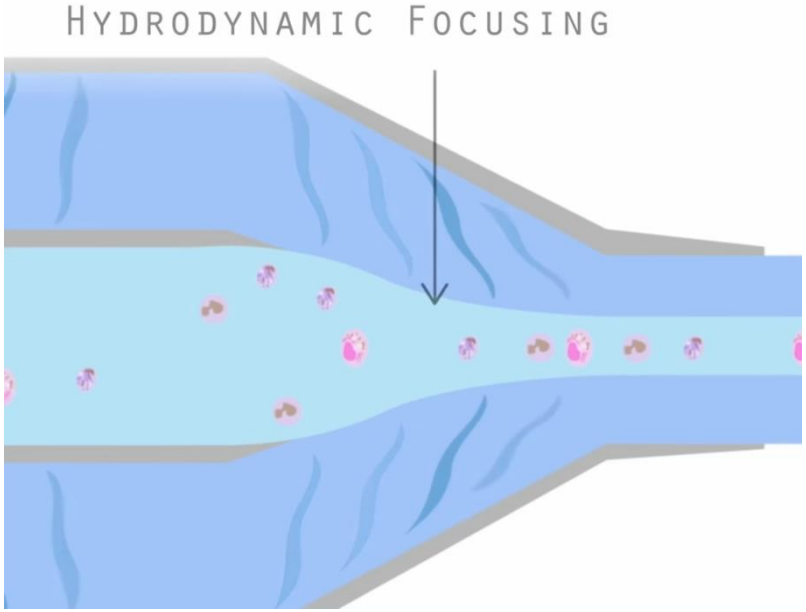


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Fluorescence Activated Cell Sorting

Fluorescence Activated Cell Sorting
Fluidics system

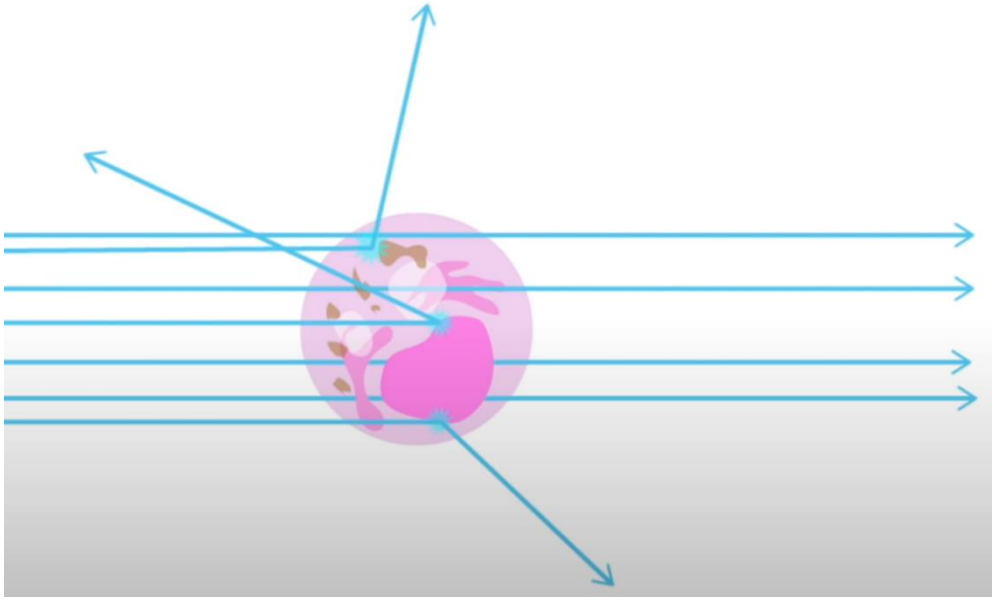




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Fluorescence Activated Cell Sorting

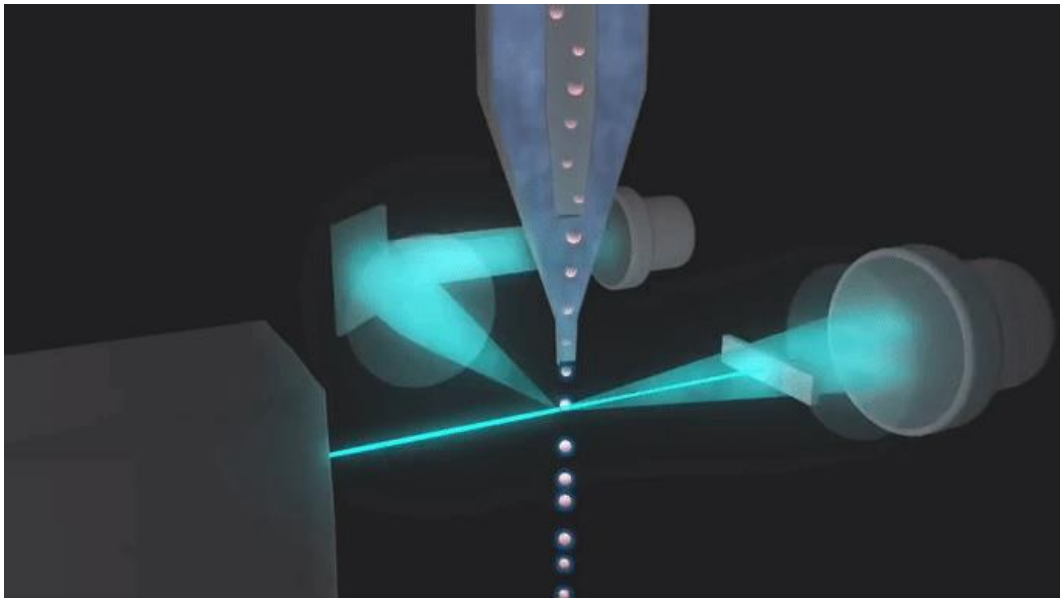
Fluorescence Activated Cell Sorting
Optical system



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Fluorescence Activated Cell Sorting

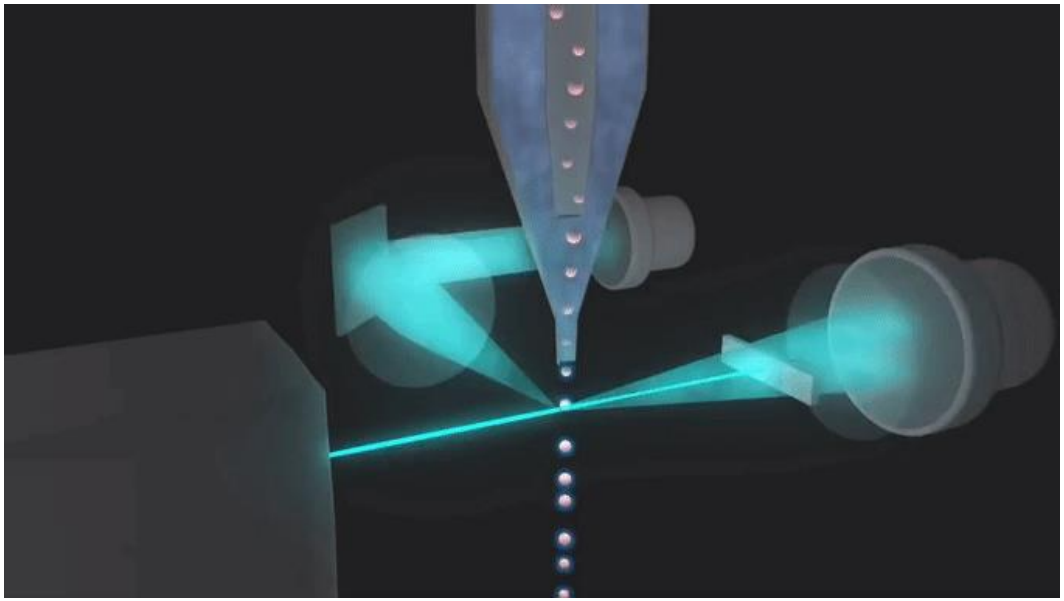
Fluorescence Activated Cell Sorting
Optical system



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Fluorescence Activated Cell Sorting

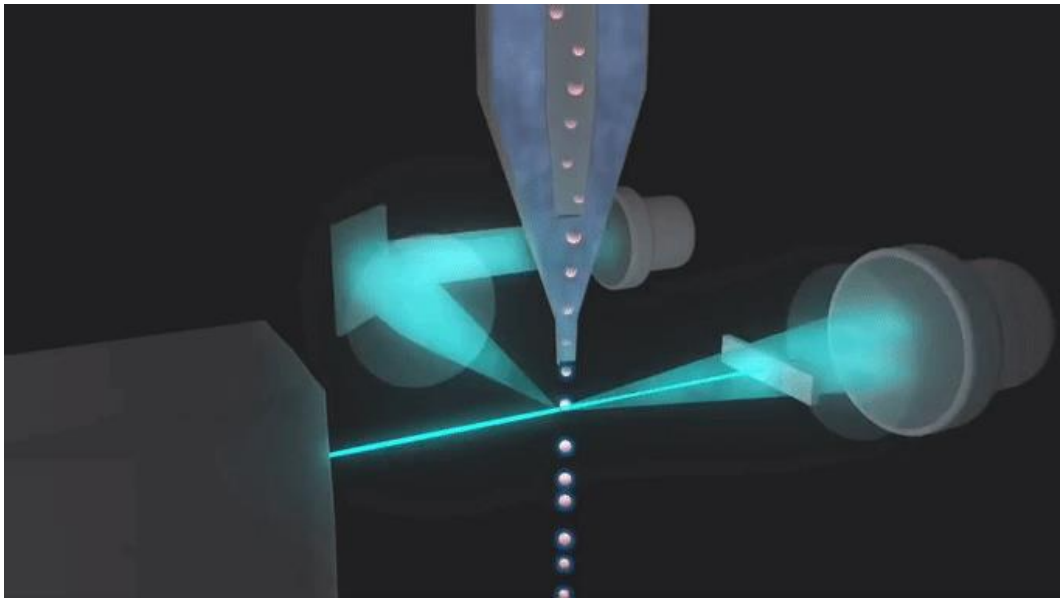
Fluorescence Activated Cell Sorting
Optical system



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Fluorescence Activated Cell Sorting

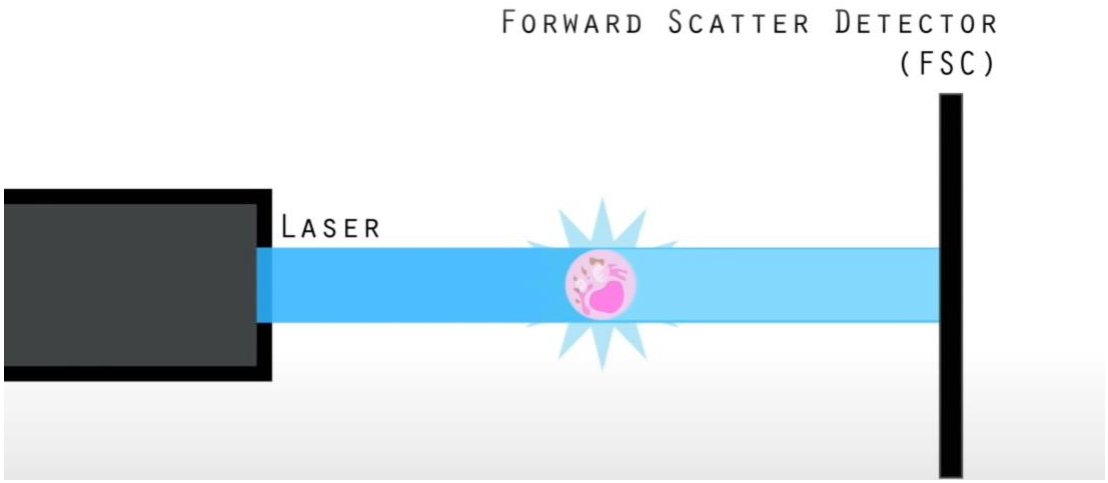
Fluorescence Activated Cell Sorting
Optical system



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Fluorescence Activated Cell Sorting

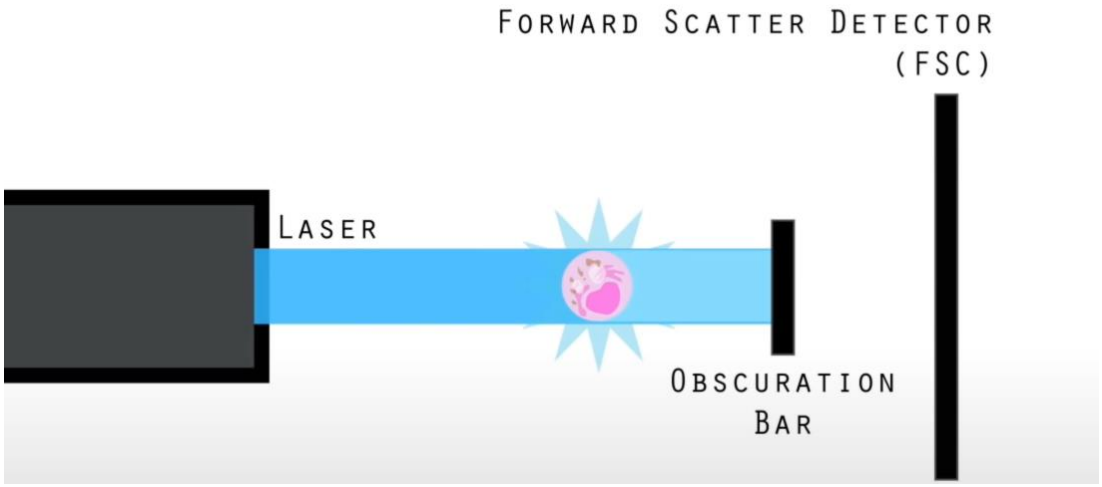
Fluorescence Activated Cell Sorting
Optical system



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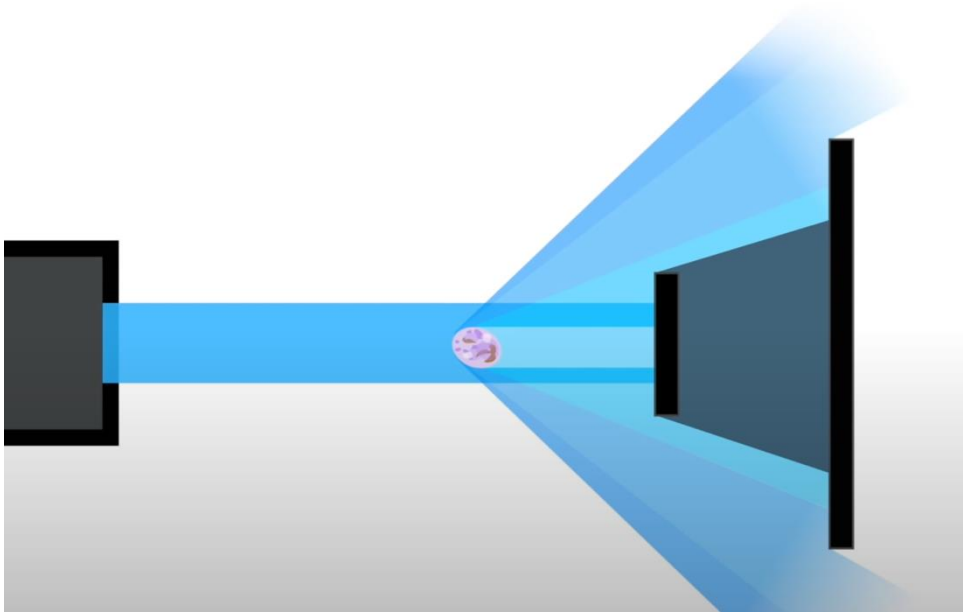
Fluorescence Activated Cell Sorting

Fluorescence Activated Cell Sorting
Optical system



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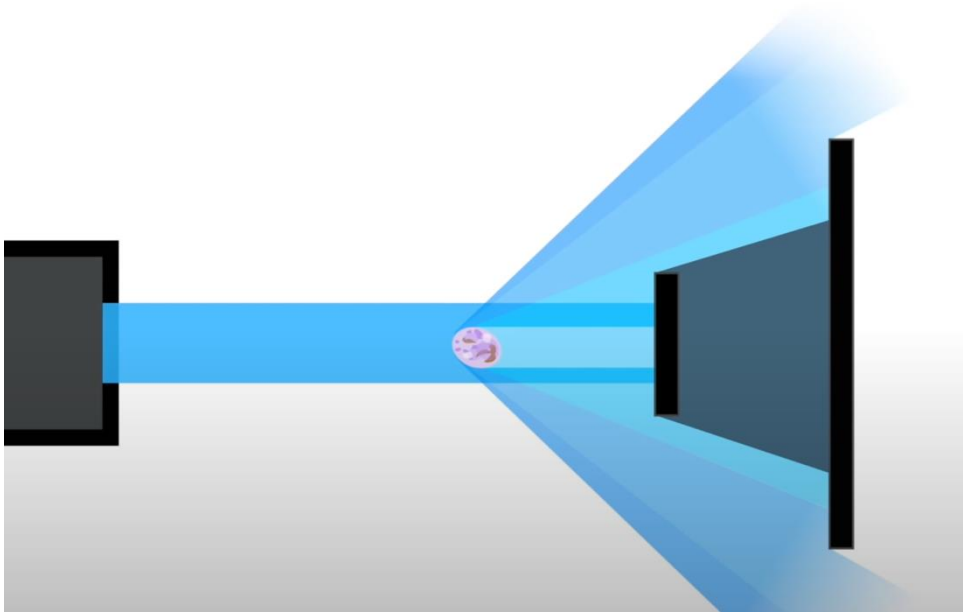
Fluorescence Activated Cell Sorting
Optical system



What does it depend on?

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Fluorescence Activated Cell Sorting
Optical system



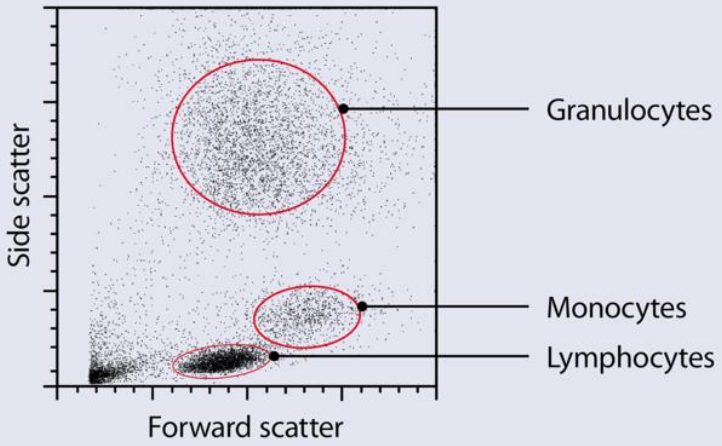
What does it depend on?

- Optical density of the cells
- Surface roughness
- Size

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Fluorescence Activated Cell Sorting

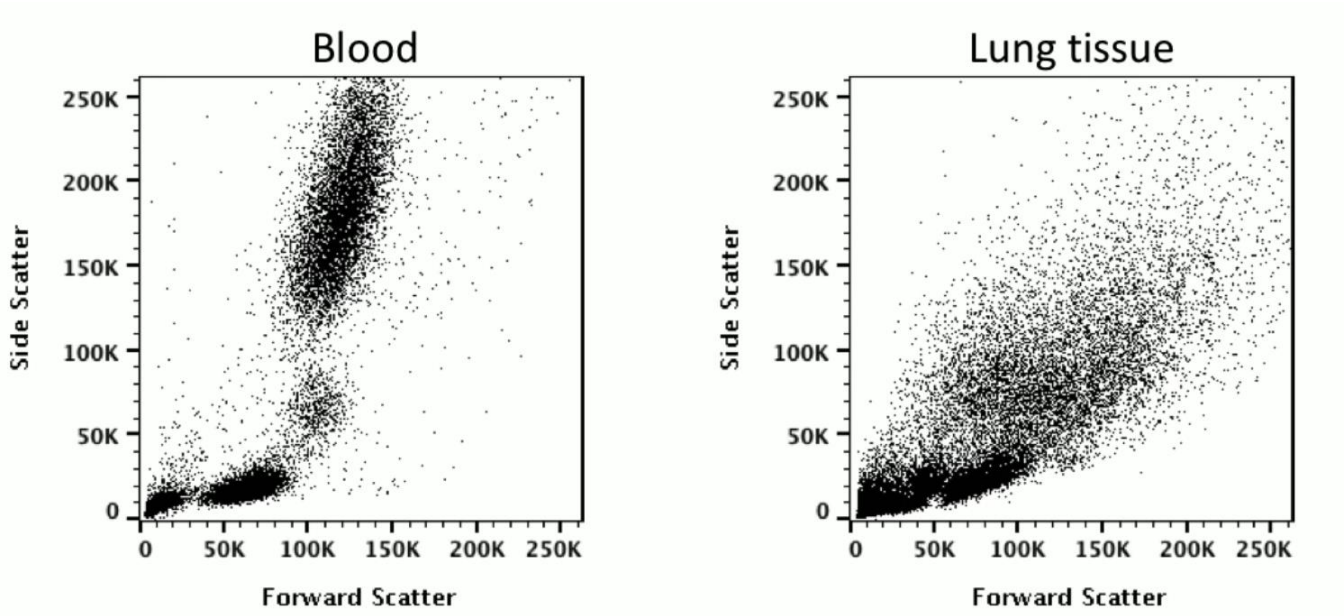
Blood sample



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Fluorescence Activated Cell Sorting

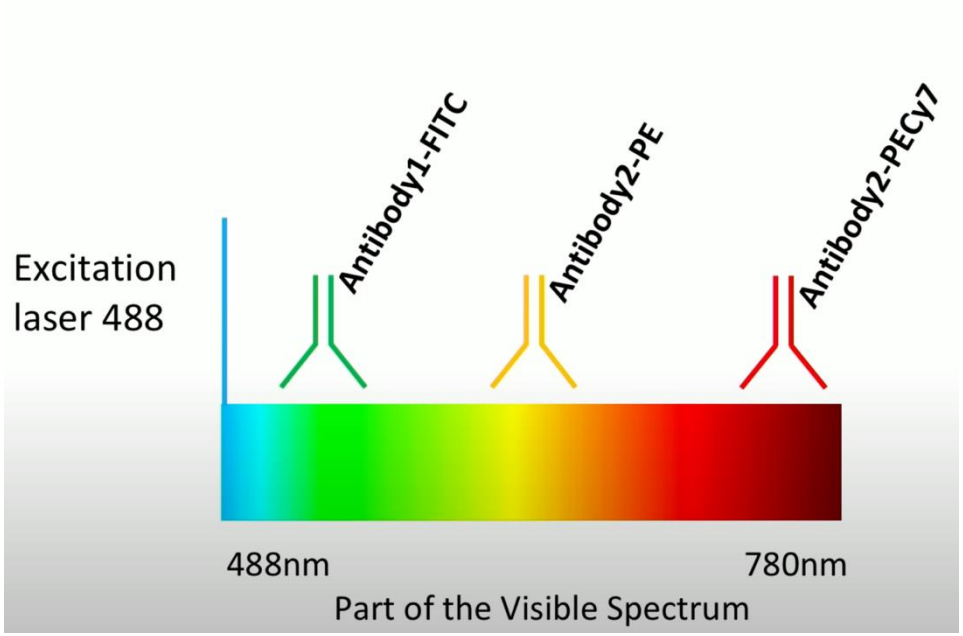
Fluorescence Activated Cell Sorting



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Fluorescence Activated Cell Sorting

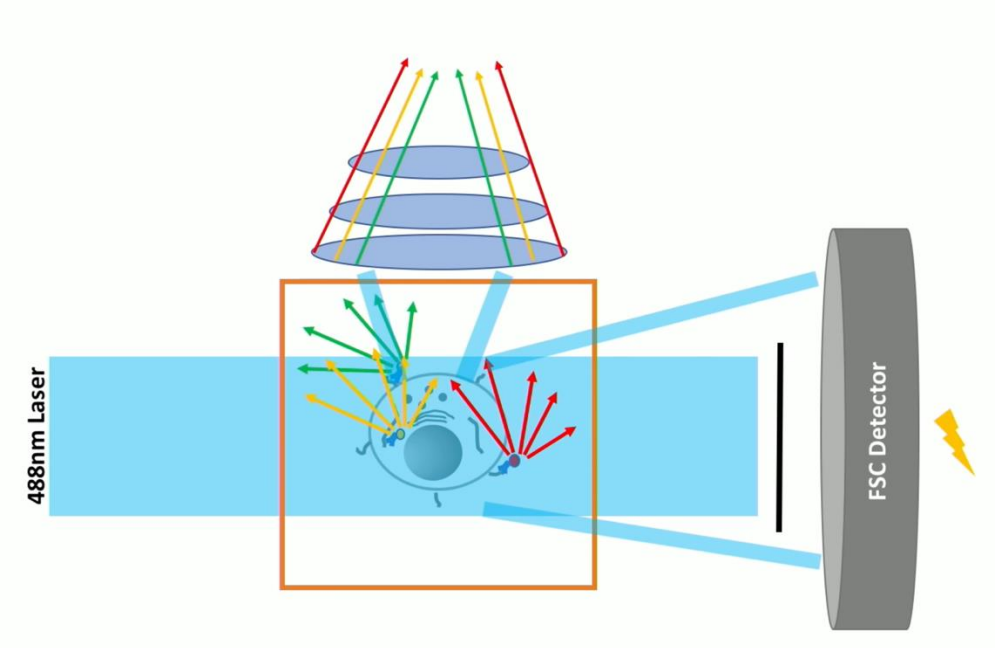
Fluorescence Activated Cell Sorting



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Fluorescence Activated Cell Sorting

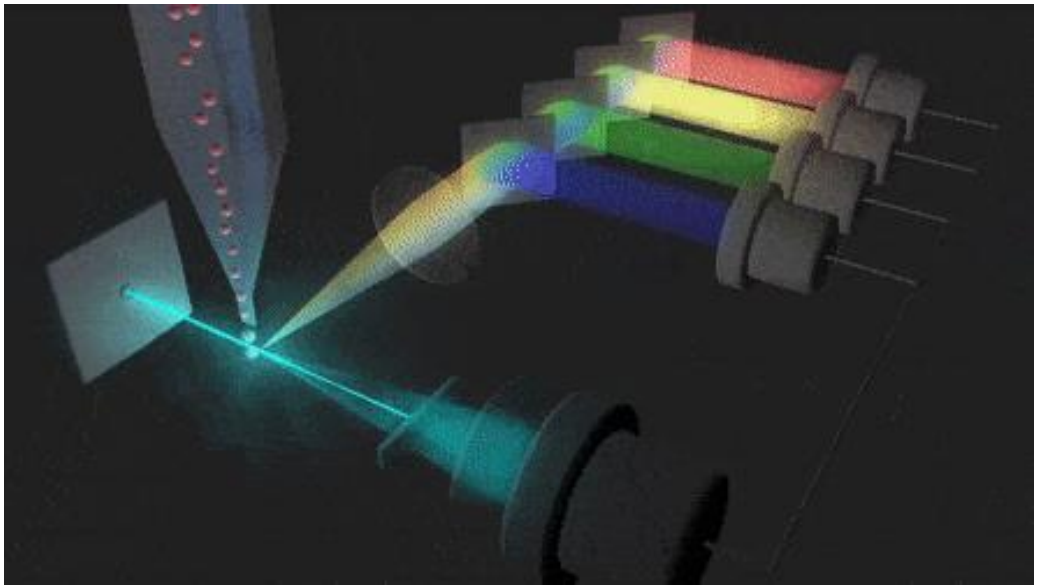
Fluorescence Activated Cell Sorting



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Fluorescence Activated Cell Sorting

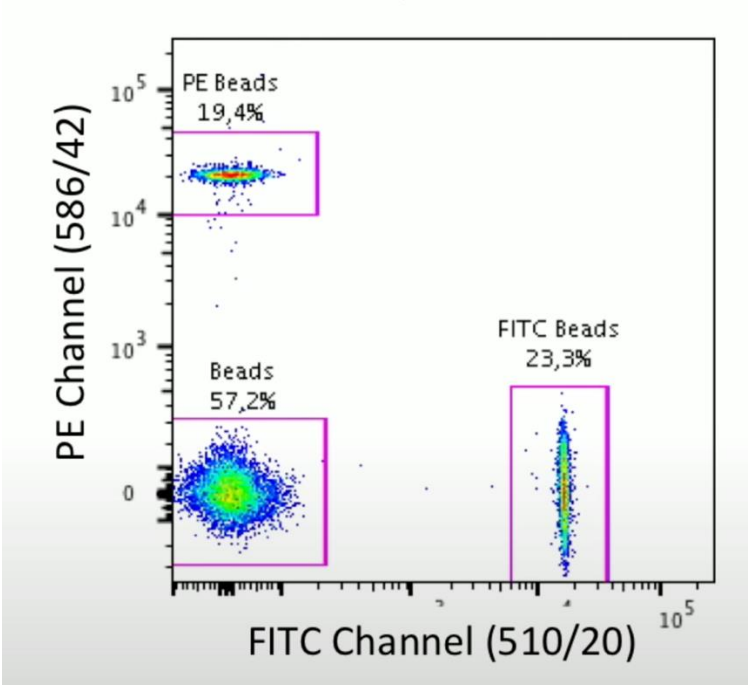
Fluorescence Activated Cell Sorting



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Fluorescence Activated Cell Sorting



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Fluorescence Activated Cell Sorting

Millikan experiment:

